14

Fluid Mechanics

CHAPTER OUTLINE

14.1 Pressure

14.2 Variation of Pressure with Depth

14.3 Pressure Measurements

14.4 Buoyant Forces and Archimedes's Principle

14.5 Fluid Dynamics

14.6 Bernoulli's Equation

14.7 Other Applications of Fluid Dynamics

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ14.1 Answer (c). Both must be built the same. A dam must be constructed to withstand the pressure at the bottom of the dam. The pressure at the bottom of a dam due to water is $P = \rho gh$, where h is the height of the water. If both reservoirs are equally high (meaning the water is equally deep), the pressure is the same regardless of width.

OQ14.2 Answer (b), (e). The buoyant force on an object is equal to the weight of the volume of water displaced by that object.

OQ14.3 Answer (d), (e). The buoyant force on the block is equal to the WEIGHT of the volume of water it displaces.

OQ14.4 Answer (b). The apple does not change volume appreciably in a dunking bucket, and the water also keeps constant density. Then the buoyant force is constant at all depths.

OQ14.5 Answer (c). The water keeps nearly constant density as it increases in pressure with depth. The beach ball is compressed to smaller volume as you take it deeper, so the buoyant force decreases. Note that the

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situation this question considers is different from that of OQ14.2. In OQ14.2, the beach ball is fully inflated at a pressure higher than 1 atm, and the tension from the plastic balances the excess pressure. So even when the ball is 1 m under water, the water pressure increases, so the plastic tension decreases, but the inside pressure remains practically constant, hence no volume change.

OQ14.6 Answer (a), (c). Both spheres have the same volume, so the buoyant force is the same on each. The lead sphere weighs more, so its string tension must be greater.

OQ14.7 Answer (c). The absolute pressure at depth h below the surface of a fluid having density ρ is $P = P_0 + \rho gh,$ where P_0 is the pressure at the upper surface of that fluid. The fluid in each of the three vessels has density $\rho = \rho_{\text{water}}$, the top of each vessel is open to the atmosphere so that $P_0 = P_{\text{atm}}$ in each case, and the bottom is at the same depth h below the upper surface for the three vessels. Thus, the pressure P at the bottom of each vessel is the same.

OQ14.8 Answer (b). Ice on the continent of Antarctica is above sea level. At the north pole, the melting of the ice floating in the ocean will not raise the ocean level (see OQ14.15).

OQ14.9 Answer (c). The normal force from the bottom plus the buoyant force from the water together balance the weight of the boat.

OQ14.10 (i) Answer (b). (ii) Answer (c). When the steel is underwater, the water exerts on the steel a buoyant force that was not present when the steel was on top surrounded by air. Thus, slightly less wood will be below the water line on the wooden block. It will float higher. In both orientations the compound floating object displaces its own weight of water, so it displaces equal volumes of water. The water level in the tub will be unchanged when the object is turned over.

OQ14.11 Answer (b). The excess pressure is transmitted undiminished throughout the container. It will compress air inside the wood. The water driven into the pores of the wood raises the block's average density and makes if float lower in the water. Add some thumbtacks to reach neutral buoyancy and you can make the wood sink or rise at will by subtly squeezing a large clear–plastic soft–drink bottle. René Descartes invented this toy or trick, called a Cartesian diver.

OQ14.12 Answer (b). The level of the pond falls. This is because the anchor displaces more water while in the boat. A floating object displaces a volume of water whose weight is equal to the weight of the object. A submerged object displaces a volume of water equal to the volume of the object. Because the density of the anchor is greater than that of water, a volume of water that weighs the same as the anchor will be

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greater than the volume of the anchor.

OQ14.13 Answer: (b) = (d) = (e) > (a) > (c). Objects (a) and (c) float, and (e) barely floats (we ignore the thin-walled bottle). On them the buoyant forces are equal to the gravitational forces exerted on them, so the ranking is (e) greater than (a) and (e) greater than (c). Objects (b) and (d) sink, and have volumes equal to (e), so they feel equal-size buoyant forces: (e) = (b) = (d).

OQ14.14 Answer (d). You want the water drop-Earth system to have four times the gravitational potential energy, relative to where the water drop leaves the nozzle, as a water drop turns around at the top of the fountain. Therefore, you want it to start out with four times the kinetic energy, which means with twice the speed at the nozzle. Given the constant volume flow rate Av, you want the area to be two times smaller. If the nozzle has a circular opening, you need to decrease its radius only by the square root of two.

OQ14.15 Answer (c). The water level stays the same. The solid ice displaced its own mass of liquid water. The meltwater does the same.

OQ14.16 Answer (e). Since the pipe is horizontal, each part of it is at the same vertical level or has the same y coordinate. Thus, from Bernoulli's equation $P+\frac{1}{2}\rho v^2+\rho g y=\text{constant},$ we see that the sum of the pressure and the kinetic energy per unit volume ($P+\frac{1}{2}\rho v^2$) must also be constant throughout the pipe.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ14.1 The horizontal force exerted by the outside fluid, on an area element of the object's side wall, has equal magnitude and opposite direction to the horizontal force the fluid exerts on another element diametrically opposite the first.

CQ14.2 The weight depends upon the total volume of water in the glass. The pressure at the bottom depends only on the depth. With a cylindrical glass, the water pushes only horizontally on the side walls and does not contribute to an extra downward force above that felt by the base. On the other hand, if the glass is wide at the top with a conical shape, the water pushes outward and downward on each bit of side wall. The downward components add up to an extra downward force, more than that exerted on the small base area.

CQ14.3 The air in your lungs, the blood in your arteries and veins, and the

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protoplasm in each cell exert nearly the same pressure, so that the wall of your chest can be in equilibrium.

CQ14.4 Yes. The propulsive force of the fish on the water causes the scale reading to fluctuate. Its average value will still be equal to the total weight of bucket, water, and fish. In other words, the center of mass of the fish-water-bucket system is moving around when the fish swims. Therefore, the net force acting on the system cannot be a constant. Apart from the weights (which are constants), the vertical force from the scale is the only external force on the system: it changes as the center of mass moves (accelerates). So the scale reading changes.

CQ14.5 (a) The greater air pressure inside the spacecraft causes air to be expelled through the hole.

(b) Clap your shoe or wallet over the hole, or a seat cushion, or your hand. Anything that can sustain a force on the order of 100 N is strong enough to cover the hole and greatly slow down the escape of the cabin air. You need not worry about the air rushing out instantly, or about your body being "sucked" through the hole, or about your blood boiling or your body exploding. If the cabin pressure drops a lot, your ears will pop and the saliva in your mouth may boil—at body temperature—but you will still have a couple of minutes to plug the hole and put on your emergency oxygen mask. Passengers who have been drinking carbonated beverages may find that the carbon dioxide suddenly comes out of solution in their stomachs, distending their vests, making them belch, and all but frothing from their ears; so you might warn them of this effect.

CQ14.6 The rapidly moving air above the ball exerts less pressure than the atmospheric pressure below the ball. This can give substantial lift to balance the weight of the ball.

CQ14.7 Imagine there have been large water demands and the water vessel at the top is half full. The depth of water from the upper water surface to the ground is still large. Therefore, the pressure at the base of the water is only slightly reduced from that due to a full tank, resulting in adequate water pressure at residents' faucets. If the water tank were a tall cylinder, a half-full tank would be only half as deep and the pressure at residents' faucets would be only half as great. Also, the water level in a tall cylinder would drop faster, because its cross-sectional area is smaller, so it would have to be replaced more often.

CQ14.8 Like the ball, the balloon will remain in front of you. It will not bob up to the ceiling. Air pressure will be no higher at the floor of the

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sealed car than at the ceiling. The balloon will experience no buoyant force. You might equally well switch off gravity. In the freely falling elevator, everything is effectively "weightless," so the air does not exert a buoyant force on anything.

CQ14.9 (a) Yes. (b) Yes. (c) The buoyant force is a conservative force. It does positive work on an object moving upward in a fluid and an equal amount of negative work on the object moving down between the same two elevations. [Note that mechanical energy, $K + U$, is not conserved here because of viscous drag from the water.] Potential energy is not associated with the object on which the buoyant force acts, but with the system of objects interacting by the buoyant force. This system is the immersed object and the fluid.

CQ14.10 The metal is more dense than water. If the metal is sufficiently thin, it can float like a ship, with the lip of the dish above the water line. Most of the volume below the water line is filled with air. The mass of the dish divided by the volume of the part below the water line is just equal to the density of water. Placing a bar of soap into this space to replace the air raises the average density of the compound object and the density can become greater than that of water. The dish sinks with its cargo.

CQ14.11 Use a balance to determine its mass. Then partially fill a graduated cylinder with water. Immerse the rock in the water and determine the volume of water displaced. Divide the mass by the volume and you have the density. It may be more precise to hang the rock from a string, measure the force required to support it under water, and subtract to find the buoyant force. The buoyant force can be thought of as the weight of so many grams of water, which is that number of cubic centimeters of water, which is the volume of the submerged rock. This volume with the actual rock mass tells you its density.

CQ14.12 The diet drink fluid has no dissolved sugar, so its density is less than that of the regular drink. Try it.

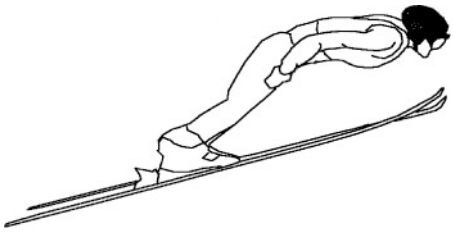
CQ14.13 At lower elevation the water pressure is greater because pressure increases with increasing depth below the water surface in the reservoir (or water tower). The penthouse apartment is not so far below the water surface. The pressure behind a closed faucet is weaker there and the flow weaker from an open faucet. Your fire department likely has a record of the precise elevation of every fire hydrant.

CQ14.14 The boat floats higher in the ocean than in the inland lake. According to Archimedes's principle, the magnitude of buoyant force on the ship is equal to the weight of the water displaced by the ship. Because the density of salty ocean water is greater than fresh lake

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water, less ocean water needs to be displaced to enable the ship to float.

CQ14.15 The ski jumper gives her body the shape of an airfoil. She deflects the air stream downward as it rushes past and the airstream deflects her upward by Newton's third law. The air exerts on her a lift force, giving her a higher and longer trajectory.



ANS FIG. CQ14.15

CQ14.16 When taking off into the wind, the increased airspeed over the wings gives a larger lifting force, enabling the pilot to take off in a shorter length of runway.

CQ14.17 A breeze from any direction speeds up to go over the mound and the air pressure drops. Air then flows through the burrow from the lower entrance to the upper entrance.

CQ14.18 (a) Since the velocity of the air in the right-hand section of the pipe is lower than that in the middle, the pressure is higher.

(b) The equation that predicts the same pressure in the far right- and left-hand sections of the tube assumes laminar flow without viscosity. The equation also assumes the fluid is incompressible, but air is not. Also, the left-hand tube is open to the atmosphere while the right-hand tube is not. Internal friction will cause some loss of mechanical energy, and turbulence will also progressively reduce the pressure. If the pressure at the left were not lower than at the right, the flow would stop.

CQ14.19 The stored corn in the silo acts as a fluid: the greater the depth, the greater the pressure on the sides of the silo. The metal bands are placed closer, or doubled, at lower portions to provide more force to balance the force from the greater pressure.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 14.1 Pressure

P14.1 We shall assume that each chair leg supports one-fourth of the total weight so the normal force each leg exerts on the floor is $n = mg/4$. The pressure of each leg on the floor is then

$$P_{\text{leg}} = \frac{n}{A_{\text{leg}}} = \frac{mg/4}{\pi r^2} = \frac{(95.0\ \text{kg})(9.80\ \text{m/s}^2)}{4\pi(0.500\times10^{-2}\ \text{m})^2} = \boxed{2.96\times10^{6}\ \text{Pa}}$$

P14.2 (a) If the particles in the nucleus are closely packed with negligible space between them, the average nuclear density should be approximately that of a proton or neutron. That is

$$\rho_{\text{nucleus}} \approx \frac{m_{\text{proton}}}{V_{\text{proton}}} = \frac{m_{\text{proton}}}{4\pi r^3/3} \sim \frac{3(1.67\times10^{-27}\ \text{kg})}{4\pi(1\times10^{-15}\ \text{m})^3}$$

$$\boxed{\sim 4\times10^{17}\ \text{kg/m}^3}$$

(b) The density of an atom is about 10^{14} times greater than the density of iron and other common solids and liquids. This shows that an atom is mostly empty space. Liquids and solids, as well as gases, are mostly empty space.

P14.3 (a) $$P = \frac{F}{A} = \frac{(50.0\ \text{kg})(9.80\ \text{m/s}^2)}{\pi(0.500\times10^{-2}\ \text{m})^2} = \boxed{6.24\times10^{6}\ \text{N/m}^2}$$

(b) The pressure from the heel might damage the vinyl floor covering.

P14.4 The Earth's surface area is $4\pi R^2$. The force pushing inward over this area amounts to

$$F = P_0 A = P_0(4\pi R^2)$$

This force is the weight of the air:

$$F_g = mg = P_0(4\pi R^2)$$

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so, assuming g is everywhere the same, the mass of the air is

$$m = \frac{P_0\left(4\pi R^2\right)}{g}$$

$$= \frac{\left(1.013\times10^5\ \text{N/m}^2\right)\left[4\pi\left(6.37\times10^6\ \text{m}\right)^2\right]}{9.80\ \text{m/s}^2}$$

$$= \boxed{5.27\times10^{18}\ \text{kg}}$$

P14.5 The definition of density, $\rho = m/V$, is often most directly useful in the form $m = \rho V$.

$$V = \ell wh \quad \text{so} \quad m = \rho V = \rho \ell wh$$

Thus $$m = (19.3\times10^3\ \text{kg/m}^3)(4.50\ \text{cm})(11.0\ \text{cm})(26.0\ \text{cm})$$

$$= (19.3\times10^3\ \text{kg/m}^3)(1\ 290\ \text{cm}^3)(1\ \text{m}^3/10^6\ \text{cm}^3) = \boxed{24.8\ \text{kg}}$$

Section 15.2 Variation of Pressure with Depth

P14.6 (a) Suppose the "vacuum cleaner" functions as a high–vacuum pump. The air below the brick will exert on it a lifting force

$$F = PA = \left(1.013\times10^5\ \text{Pa}\right)\left[\pi\left(1.43\times10^{-2}\ \text{m}\right)^2\right] = \boxed{65.1\ \text{N}}$$

(b) The octopus can pull the bottom away from the top shell with a force that could be no larger than

$$F = PA = \left(P_0 + \rho gh\right)A$$

$$= \left[1.013\times10^5\ \text{Pa} + \left(1\ 030\ \text{kg/m}^3\right)\left(9.80\ \text{m/s}^2\right)\left(32.3\ \text{m}\right)\right]$$

$$\times\left[\pi\left(1.43\times10^{-2}\ \text{m}\right)^2\right]$$

$$F = \boxed{275\ \text{N}}$$

P14.7 Assuming the spring obeys Hooke's law, the increase in force on the piston required to compress the spring an additional amount Δx is

$$\Delta F = F - F_0 = \left(P - P_0\right)A = k\left(\Delta x\right)$$

The gauge pressure at depth h beneath the surface of a fluid is

$$P - P_0 = \rho gh$$

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so we have

$$\rho g h A = k(\Delta x)$$

or the required depth is

$$h = k(\Delta x)/\rho g A$$

If $k = 1\,250$ N/m, $A = \pi d^2/4$, $d = 1.20 \times 10^{-2}$ m, and the fluid is water ($\rho = 1.00 \times 10^3$ kg/m^3), the depth required to compress the spring an additional $\Delta x = 0.750 \times 10^{-2}$ m is

$$h = \boxed{8.46 \text{ m}}$$

P14.8 Since the pressure is the same on both sides, $\frac{F_1}{A_1} = \frac{F_2}{A_2}$, and

in this case, $\frac{15\,000 \text{ N}}{200 \text{ cm}^2} = \frac{F_2}{3.00 \text{ cm}^2}$ or $F_2 = \boxed{225 \text{ N}}$

P14.9 $F_g = (80.0 \text{ kg})(9.80 \text{ m/s}^2) = 784 \text{ N}$

When the cup barely supports the student, the normal force of the ceiling is zero and the cup is in equilibrium.

$$F_g = F = PA = (1.013 \times 10^5 \text{ Pa})A$$

$$A = \frac{F_g}{P} = \frac{784 \text{ N}}{1.013 \times 10^5 \text{ Pa}} = \boxed{7.74 \times 10^{-3} \text{ m}^2}$$

P14.10 The pressure on the bottom due to the water is $P_b = \rho g z = 1.96 \times 10^4$ Pa.

(a) The force exerted by the water on the bottom is then

$$F_b = P_b A = (1.96 \times 10^4 \text{ Pa})(30.0 \text{ m})(10.0 \text{ m})$$

$$= \boxed{5.88 \times 10^6 \text{ N down}}$$

Pressure varies with depth. On a strip of height dz and length L, the force is $dF = PdA = PLdz = \rho g z L dz$, which gives the integral

$$F = \int_0^h \rho g z L dz = \frac{1}{2}\rho g L h^2 = \left(\frac{1}{2}\rho g h\right) L h = P_{\text{average}} A$$

(b) On each end,

$$F = P_{\text{average}} A = (9.80 \times 10^3 \text{ Pa})(20.0 \text{ m}^2) = \boxed{196 \text{ kN outward}}$$

(c) On the side,

$$F = P_{\text{average}} A = (9.80 \times 10^3 \text{ Pa})(60.0 \text{ m}^2) = \boxed{588 \text{ kN outward}}$$

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P14.11 (a) At a depth of 27.5 m, the absolute pressure is

$$P = P_0 + \rho g h = 101.3 \times 10^3 \text{ Pa} + \left(1.00 \times 10^3 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)(27.5 \text{ m})$$

$$= \boxed{3.71 \times 10^5 \text{ Pa}}$$

(b) The inward force the water will exert on the window is

$$F = PA = P\left(\pi r^2\right) = \left(3.71 \times 10^5 \text{ Pa}\right)\pi\left(\frac{35.0 \times 10^{-2} \text{ m}}{2}\right)^2$$

$$= \boxed{3.57 \times 10^4 \text{ N}}$$

P14.12 We imagine Superman can produce a perfect vacuum in the straw. Take point 1, at position $y_1 = 0$, to be at the water's surface and point 2, at position y_2 = length of straw, to be at the upper end of the straw. What is the greatest length of straw that will allow Superman to drink? Solve for y_2:

$$P_1 + \rho g y_1 = P_2 + \rho g y_2$$

$$1.013 \times 10^5 \text{ Pa} + 0 = 0 + (10^3 \text{ kg/m}^3)(9.80 \text{ m/s}^2) y_2$$

or $y_2 = 10.3$ m.

The situation is impossible because the longest straw Superman can use and still get a drink is less than 12.0 m.

***P14.13** The excess water pressure (over air pressure) halfway down is

$$P_{\text{gauge}} = \rho g h = \left(1\,000 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)(1.20 \text{ m})$$

$$= 1.18 \times 10^4 \text{ Pa}$$

The force on the wall due to the water is

$$F = P_{\text{gauge}} A = \left(1.18 \times 10^4 \text{ Pa}\right)(2.40 \text{ m})(9.60 \text{ m})$$

$$= \boxed{2.71 \times 10^5 \text{ N}}$$

horizontally toward the back of the hole.

***P14.14** We first find the absolute pressure at the interface between oil and water:

$$P_1 = P_0 + \rho_{\text{oil}} g h_{\text{oil}}$$

$$= 1.013 \times 10^5 \text{ Pa} + \left(700 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)(0.300 \text{ m})$$

$$= 1.03 \times 10^5 \text{ Pa}$$

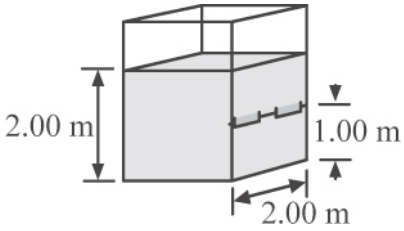
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This is the pressure at the top of the water. To find the absolute pressure at the bottom, we use

$$P_2 = P_1 + \rho_{\text{water}} g h_{\text{water}}$$
$$= 1.03 \times 10^5 \text{ Pa} + \left(10^3 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)(0.200 \text{ m})$$
$$= \boxed{1.05 \times 10^5 \text{ Pa}}$$

P14.15 The air outside and water inside both exert atmospheric pressure, so only the excess water pressure ρgh counts for the net force. Take a strip of hatch between depth h and $h + dh$. It feels force

$$dF = PdA = \rho gh(2.00 \text{ m})dh$$



ANS. FIG. P14.15

(a) The total force is

$$F = \int dF = \int_{h=1.00 \text{ m}}^{2.00 \text{ m}} \rho gh(2.00 \text{ m})dh$$

$$F = \rho g(2.00 \text{ m})\frac{h^2}{2}\bigg|_{1.00 \text{ m}}^{2.00 \text{ m}} = \left(1\,000 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)\frac{(2.00 \text{ m})}{2}$$
$$\times\left[(2.00 \text{ m})^2 - (1.00 \text{ m})^2\right]$$

$$F = \boxed{29.4 \text{ kN (to the right)}}$$

(b) The lever arm of dF is the distance $(h - 1.00 \text{ m})$ from hinge to strip:

$$\tau = \int d\tau = \int_{h=1.00 \text{ m}}^{2.00 \text{ m}} \rho gh(2.00 \text{ m})(h - 1.00 \text{ m})dh$$

$$\tau = \rho g(2.00 \text{ m})\left[\frac{h^3}{3} - (1.00 \text{ m})\frac{h^2}{2}\right]_{1.00 \text{ m}}^{2.00 \text{ m}}$$

$$\tau = \left(1\,000 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)(2.00 \text{ m})\left(\frac{7.00 \text{ m}^3}{3} - \frac{3.00 \text{ m}^3}{2}\right)$$

$$\tau = \boxed{16.3 \text{ kN} \cdot \text{m counterclockwise}}$$

P14.16 The air outside and water inside both exert atmospheric pressure, so only the excess water pressure ρgh counts for the net force.

(a) At a distance y from the top of the water, take a strip of hatch between depth y and $y + dy$. It feels force

$$dF = PdA = Pwdy = (\rho gyw)dy$$

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The total force is

$$F = \int_{d-h}^{d} \rho g w y dy = \frac{1}{2}\rho g w y^2 \Big|_{d-h}^{d} = \frac{1}{2}\rho g w \left[d^2 - (d-h)^2 \right]$$

$$\boxed{F = \frac{1}{2}\rho g w h (2d - h)}$$

(b) The lever arm of dF is the distance $[y - (d - h)]$ from hinge to strip:

$$\tau = \int_{d-h}^{d} \rho g w y \left[y - (d-h) \right] dy = \rho g w \int_{d-h}^{d} \left[y^2 - y(d-h) \right] dy$$

$$= \rho g w \left[\frac{y^3}{3} - \frac{(d-h)y^2}{2} \right]_{d-h}^{d}$$

$$= \frac{\rho g w}{6} \left[2d^3 - 2(d-h)^3 - 3(d-h)d^2 + 3(d-h)^3 \right]$$

$$= \frac{\rho g w}{6} \left[2d^3 - 3(d-h)d^2 + (d-h)^3 \right]$$

$$= \frac{\rho g w}{6} \left[2d^3 - 3d^3 + 3d^2 h + d^3 - 3d^2 h + 3dh^2 - h^3 \right]$$

$$\tau = \frac{\rho g w}{6} \left[+3dh^2 - h^3 \right]$$

$$\boxed{\tau = \frac{1}{2}\rho g w \left(dh^2 - \frac{1}{3}h^3 \right)}$$

***P14.17** The fluid in the hydraulic jack is originally exerting the same pressure as the air outside. This pressure P_0 results in zero net force on either piston. For the equilibrium of piston 2 we require

$$500 \text{ lb} = (P - P_0)A = (P - P_0)\pi \left(\frac{1.50 \text{ in.}}{2} \right)^2$$

Let F_1 represent the force the lever bar exerts on piston 1. Then similarly

$$F_1 = (P - P_0)\pi \left(\frac{0.250 \text{ in.}}{2} \right)^2$$

We ignore the weights of the pistons, sliding friction, and the slight difference in fluid pressure P due to the height difference between points 1 and 2. By division,

$$\frac{F_1}{500 \text{ lb}} = \left(\frac{0.250 \text{ in.}}{1.50 \text{ in.}} \right)^2 \quad \rightarrow \quad F_1 = \frac{500 \text{ lb}}{36.0}$$

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We say the hydraulic lift has an ideal mechanical advantage of 36. Next for the lever bar we ignore weight and friction, assume equilibrium, and take torques about the fixed hinge.

$$\sum \tau = 0 \text{ gives } F_1(2.00 \text{ in.}) - F(12.0 \text{ in.}) = 0 \text{, or } F = \frac{F_1}{6}.$$

The lever has an ideal mechanical advantage of 6. By substitution,

$$F = \frac{500 \text{ lb}}{36 \cdot 6} = \boxed{2.31 \text{ lb}}$$

P14.18 The bell is uniformly compressed, so we can model it with any shape. We choose a sphere of diameter 3.00 m.

The pressure on the ball is given by $P = P_{\text{atm}} + \rho_w g h,$ so the change in pressure on the ball from when it is on the surface of the ocean to when it is at the bottom of the ocean is $\Delta P = \rho_w g h.$

In addition,

$$\Delta V = \frac{-V\Delta P}{B} = \frac{-\rho_w g h V}{B} = -\frac{4\pi\rho_w g h r^3}{3B}$$

where B is the bulk modulus. Substituting,

$$\Delta V = -\frac{4\pi\left(1\ 030 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)(1\ 000 \text{ m})(1.50 \text{ m})^3}{3\left(14.0\times10^{10} \text{ Pa}\right)}$$

$$\Delta V = -1.02\times10^{-3} \text{ m}^3$$

From $V = \frac{4}{3}\pi r^3 \rightarrow dV = 4\pi r^2 dr,$ we use r = 1.50 m, set $dV = \Delta V$, and solve for dr:

$$dr = -3.60 \times 10^{-5} \text{ m}$$

Therefore, the diameter decreases by $\boxed{0.072\ 1 \text{ mm.}}$

Section 14.3 Pressure Measurements

P14.19 A drop of 20.0 mm of mercury is a pressure change of

$$\Delta P = \rho g \Delta h = \left(13.6\times10^3 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)\left(-20.0\times10^{-3} \text{ m}\right)$$
$$= -2.66\times10^3 \text{ Pa}$$

$$P = P_0 + \Delta P_0 = (1.013 - 0.026\ 6)\times10^5 \text{ Pa} = \boxed{0.986\times10^5 \text{ Pa}}$$

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P14.20 (a) $P = P_0 + \rho gh$ and the gauge pressure is

$$P - P_0 = \rho gh = (1\ 000\ \text{kg})(9.8\ \text{m/s}^2)(0.160\ \text{m})$$

$$= \boxed{1.57\ \text{kPa}} = (1.57 \times 10^3\ \text{Pa})\left(\frac{1\ \text{atm}}{1.013 \times 10^5\ \text{Pa}}\right)$$

$$= \boxed{0.015\ 5\ \text{atm}}$$

It would lift a mercury column to height

$$h = \frac{P - P_0}{\rho g} = \frac{1\ 568\ \text{Pa}}{(13\ 600\ \text{kg/m}^3)(9.80\ \text{m/s}^2)} = \boxed{11.8\ \text{mm}}$$

(b) Blockage of the fluid within the spinal column or between the skull and the spinal column would prevent the fluid level from rising.

P14.21 (a) To find the height of the column of wine, we use

$$P_0 = \rho gh$$

then

$$h = \frac{P_0}{\rho g}$$

$$= \frac{1.013 \times 10^5\ \text{Pa}}{(0.984 \times 10^3\ \text{kg/m}^3)(9.80\ \text{m/s}^2)}$$

$$= \boxed{10.5\ \text{m}}$$



ANS. FIG. P14.21

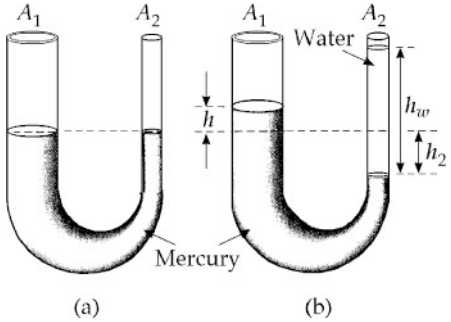
(b) No. The vacuum is not as good because some alcohol and water will evaporate. The equilibrium vapor pressures of alcohol and water are higher than the vapor pressure of mercury.

P14.22 (a) Using the definition of density, we have

$$h_w = \frac{m_{\text{water}}}{A_2 \rho_{\text{water}}}$$

$$= \frac{100\ \text{g}}{(5.00\ \text{cm}^2)(1.00\ \text{g/cm}^3)}$$

$$= \boxed{20.0\ \text{cm}}$$



ANS. FIG. P14.22

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(b) ANS. FIG. P14.22 (b) represents the situation after the water is added. A volume $\left(A_2 h_2\right)$ of mercury has been displaced by water in the right tube. The additional volume of mercury now in the left tube is $A_1 h$. Since the total volume of mercury has not changed,

$$A_2 h_2 = A_1 h \quad \text{or} \quad h_2 = \frac{A_1}{A_2} h \qquad \textbf{[1]}$$

At the level of the mercury–water interface in the right tube, we may write the absolute pressure as:

$$P = P_0 + \rho_{\text{water}} g h_w$$

The pressure at this same level in the left tube is given by

$$P = P_0 + \rho_{\text{Hg}} g\left(h + h_2\right) = P_0 + \rho_{\text{water}} g h_w$$

which, using equation [1] above, reduces to

$$\rho_{\text{Hg}} h\left[1 + \frac{A_1}{A_2}\right] = \rho_{\text{water}} h_w$$

or $$h = \frac{\rho_{\text{water}} h_w}{\rho_{\text{Hg}}\left(1 + A_1/A_2\right)}.$$

Thus, the level of mercury has risen a distance of

$$h = \frac{\left(1.00\ \text{g/cm}^3\right)(20.0\ \text{cm})}{\left(13.6\ \text{g/cm}^3\right)(1 + 10.0/5.00)}$$

$\boxed{h = 0.490\ \text{cm}}$ above the original level.

P14.23 (a) We can directly write the bottom pressure as $P = P_0 + \rho g h$, or we can say that the bottom of the tank must support the weight of the water:

$$PA - P_0 A = m_{\text{water}} g = \rho V g = \rho A h g$$

which gives again

$$P = P_0 + \rho g h$$

The absolute pressure at depth h = 1.50 m is

$$P = P_0 + \rho g h = 101.3\ \text{kPa} + (1\,000\ \text{kg/m}^3)(9.80\ \text{m/s}^2)(1.50\ \text{m})$$

$$= \boxed{116\ \text{kPa}}$$

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(b) Now the bottom of the tank must support the weight of the whole contents. Before the people enter, P = 116 kPa. Afterwards,

$$\Delta P = \frac{Mg}{A} = \frac{(150\text{ kg})(9.80\text{ m/s}^2)}{\pi(3.00\text{ m})^2} = \boxed{52.0\text{ Pa}}$$

P14.24 (a) We can directly write the bottom pressure as $P = P_0 + \rho gh$, or we can say that the bottom of the tank must support the weight of the water:

$$PA - P_0A = m_{\text{water}}g = \rho Vg = \rho Ahg$$

which gives again

$$\boxed{P = P_0 + \rho gh}$$

(b) Now, the bottom of the tank must support the weight of the whole contents:

$$P_bA - P_0A = m_{\text{water}}g + Mg = \rho Vg + Mg = \rho Ahg + Mg$$

and this gives

$$P_b = P_0 + \rho hg + Mg/A$$

Then

$$\Delta P = P_b - P = \boxed{\frac{Mg}{A}}$$

Section 14.4 Buoyant Forces and Archimedes's Principle

P14.25 At equilibrium $\sum F = 0$ or $F_{\text{app}} + mg = B$,

where B is the buoyant force.

The applied force is $F_{\text{app}} = B - mg$,

where $B = V(\rho_{\text{water}})g$

and $m = V\rho_{\text{ball}}$

So, $F_{\text{app}} = Vg(\rho_{\text{water}} - \rho_{\text{ball}}) = \frac{4}{3}\pi r^3 g(\rho_{\text{water}} - \rho_{\text{ball}})$:

$$F_{\text{app}} = \frac{4}{3}\pi(1.90\times10^{-2}\text{ m})^3(9.80\text{ m/s}^2)(10^3\text{ kg/m}^3 - 84.0\text{ kg/m}^3)$$

$$= \boxed{0.258\text{ N down}}$$

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P14.26 Refer to Figure P14.26. We observe from the left-hand diagram,

$$\sum F_y = 0 \quad \rightarrow \quad T_1 = F_g = m_{\text{object}} g = \rho_{\text{object}} g V_{\text{object}}$$

and from the right-hand diagram,

$$\sum F_y = 0 \quad \rightarrow \quad T_2 + B = F_g \quad \rightarrow \quad T_2 + B = T_1$$

which gives

$$T_2 - T_1 = B$$

where the buoyant force is

$$B = m_{\text{water}} g = \rho_{\text{w}} V_{\text{object}} g$$

Now the density of the object is

$$\rho_{\text{object}} = \frac{m_{\text{object}}}{V_{\text{object}}} = \frac{T_1/g}{B/(\rho_w g)} = \frac{\rho_w T_1}{B}$$

$$\rho_{\text{object}} = \frac{\rho_w T_1}{T_1 - T_2} = \frac{(1\ 000\ \text{kg/m}^3)(5.00\ \text{N})}{1.50\ \text{N}} = \boxed{3.33 \times 10^3\ \text{kg/m}^3}$$

P14.27 (a) We start with $P = P_0 + \rho g h$.

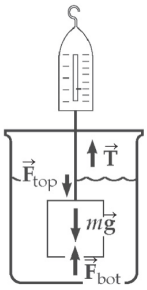
Taking $P_0 = 1.013 \times 10^5\ \text{N/m}^2$,

$\rho_{\text{water}} = 1\ 000\ \text{kg/m}^3$, and $h = 5.00\ \text{cm}$,

we find $P_{\text{top}} = 1.017\ 9 \times 10^5\ \text{N/m}^2$.

For $h = 17.0\ \text{cm}$, we get

$$P_{\text{bot}} = 1.029\ 7 \times 10^5\ \text{N/m}^2$$



ANS. FIG. P14.27

Since the areas of the top and bottom are

$$A = (0.100\ \text{m})^2 = 10^{-2}\ \text{m}^2$$

we find

$$F_{\text{top}} = P_{\text{top}} A = \boxed{1.017\ 9 \times 10^3\ \text{N}}$$

and $F_{\text{bot}} = \boxed{1.029\ 7 \times 10^3\ \text{N}}$.

(b) The tension in the string is the scale reading:

$$T = Mg - B$$

where

$$B = \rho_w V g = (10^3\ \text{kg/m}^3)(1.20 \times 10^{-3}\ \text{m}^3)(9.80\ \text{m/s}^2) = 11.8\ \text{N}$$

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and

$$Mg = (10.0\text{ kg})(9.80\text{ m/s}^2) = 98.0\text{ N}$$

Therefore,

$$T = Mg - B = 98.0\text{ N} - 11.8\text{ N} = \boxed{86.2\text{ N}}$$

(c) $F_{\text{bot}} - F_{\text{top}} = (1.0297 - 1.0179) \times 10^3\text{ N} = \boxed{11.8\text{ N}}$

which is equal to B found in part (b).

P14.28 (a) The balloon is nearly in equilibrium:

$$\sum F_y = ma_y \Rightarrow B - (F_g)_{\text{helium}} - (F_g)_{\text{payload}} = 0$$

or $$\rho_{\text{air}} gV - \rho_{\text{helium}} gV - m_{\text{payload}} g = 0$$

This reduces to

$$m_{\text{payload}} = (\rho_{\text{air}} - \rho_{\text{helium}})V$$
$$= (1.29\text{ kg/m}^3 - 0.179\text{ kg/m}^3)(400\text{ m}^3)$$
$$m_{\text{payload}} = \boxed{444\text{ kg}}$$

(b) Similarly,

$$m_{\text{payload}} = (\rho_{\text{air}} - \rho_{\text{hydrogen}})V$$
$$= (1.29\text{ kg/m}^3 - 0.089\,9\text{ kg/m}^3)(400\text{ m}^3)$$
$$m_{\text{payload}} = \boxed{480\text{ kg}}$$

The surrounding air does the lifting, nearly the same for the two balloons.

P14.29 (a) The cube has sides of length L. When floating, the horizontal top surface lies a distance h above the water's surface. The buoyant force supports the weight of the block:

$$B = \rho_{\text{water}} V_{\text{object}} g = \rho_{\text{water}} L^2 (L - h) g = \rho_{\text{wood}} L^3 g$$

Solve for h:

$$h = L - L(\rho_{\text{wood}} / \rho_{\text{water}}) = L(1 - \rho_{\text{wood}} / \rho_{\text{water}})$$
$$= (20.0\text{ cm})(1 - 0.650) = \boxed{7.00\text{ cm}}$$

(b) The buoyant force supports the weight of both blocks:

$$B = F_g + Mg, \text{ where } M = \text{mass of lead}$$

$$\rho_{\text{water}} L^3 g = \rho_{\text{wood}} L^3 g + Mg \quad \rightarrow \quad M = (\rho_{\text{water}} - \rho_{\text{wood}})L^3$$

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$$M = (1.00\ \text{kg/m}^3 - 0.650\ \text{kg/m}^3)(20.0\ \text{m})^3 = \boxed{2.80\ \text{kg}}$$

P14.30 By Archimedes's principle, the weight of the 50 planes is equal to the weight of a horizontal slice of water 11.0 cm thick and circumscribed by the water line:

$$\Delta B = \rho_{\text{water}} g(\Delta V)$$
$$50\left(2.90 \times 10^4\ \text{kg}\right) g = \left(1030\ \text{kg/m}^3\right) g\left(0.110\ \text{m}\right) A$$

giving $A = \boxed{1.28 \times 10^4\ \text{m}^2}$. The acceleration of gravity does not affect the answer.

P14.31 (a) The buoyant force of glycerin supports the weight of the sphere which is supported by the buoyant force of water.

$$B = \rho_{\text{glycerin}}\left(0.40V\right) = \rho_{\text{water}} \frac{V}{2}$$

$$\boxed{\rho_{\text{glycerin}} = \frac{\rho_{\text{water}}}{2(0.40)} = \frac{1\ 000\ \text{kg/m}^3}{0.80} = 1\ 250\ \text{kg/m}^3}$$

(b) The buoyant force from the water supports the weight of the sphere:

$$B = F_g$$

$$B = \rho_{\text{water}} \frac{V}{2} = \rho_{\text{sphere}} V$$

$$\boxed{\rho_{\text{sphere}} = \frac{\rho_{\text{water}}}{2} = 500\ \text{kg/m}^3}$$

P14.32 Constant velocity implies zero acceleration, which means that the submersible is in equilibrium under the gravitational force, the upward buoyant force, and the upward resistance force:

$$\sum F_y = ma_y = 0:$$
$$-\left(1.20 \times 10^4\ \text{kg} + m\right) g + \rho_w g V + 1\,100\ \text{N} = 0$$

where m is the mass of the added water and V is the sphere's volume. Substituting,

$$1.20 \times 10^4\ \text{kg} + m$$
$$= \left(1.03 \times 10^3\ \text{kg/m}^3\right)\left[\frac{4}{3}\pi(1.50\ \text{m})^3\right] + \frac{1\,100\ \text{N}}{9.80\ \text{m/s}^2}$$

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so $m = \boxed{2.67 \times 10^3 \text{ kg}}$.

***P14.33** (a) While the system floats, $B = w_{\text{total}} = w_{\text{block}} + w_{\text{steel}}$, or

$$\rho_w g V_{\text{submerged}} = \rho_b g V_b + m_{\text{steel}} g$$

When $m_{\text{steel}} = 0.310 \text{ kg}$, $V_{\text{submerged}} = V_b = 5.24 \times 10^{-4} \text{ m}^3$ giving

$$\rho_b = \frac{\rho_w V_b - m_{\text{steel}}}{V_b} = \rho_w - \frac{m_{\text{steel}}}{V_b}$$

$$= 1.00 \times 10^3 \text{ kg/m}^3 - \frac{0.310 \text{ kg}}{5.24 \times 10^{-4} \text{ m}^3}$$

$$= \boxed{408 \text{ kg/m}^3}$$

(b) If the total weight of the block + steel system is reduced, by having $m_{\text{steel}} < 0.310 \text{ kg}$, a smaller buoyant force is needed to allow the system to float in equilibrium. Thus, the block will displace a smaller volume of water and will be only partially submerged in the water.

(c) The block is fully submerged when $m_{\text{steel}} = 0.310$ kg. The mass of the steel object can increase slightly above this value without causing it and the block to sink to the bottom. As the mass of the steel object is gradually increased above 0.310 kg, the steel object begins to submerge, displacing additional water, and providing a slight increase in the buoyant force. With a density of about eight times that of water, the steel object will be able to displace approximately 0.310 kg/8 = 0.039 kg of additional water before it becomes fully submerged. At this point, the steel object will have a mass of about 0.349 kg and will be unable to displace any additional water. Any further increase in the mass of the object causes it and the block to sink to the bottom. In conclusion,

the block + steel system will sink if $m_{\text{stee}} \geq 0.350$ kg.

P14.34 (a) $\sum F_y = 0$: $B - T - F_g = 0 \rightarrow B - 15.0 \text{ N} - 10.0 \text{ N} = 0$

$$\boxed{B = 25.0 \text{ N}}$$

(b) The oil pushes horizontally inward on each side of the block.

(c) The string tension increases. The water under the block pushes up on the block more strongly than before because the water is under higher pressure due to the weight of the oil above it.

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(d) The pressure of the oil's weight on the water is $P = \rho_{oil} g h$, where h is the height of the oil. This pressure is transmitted to the bottom of the block, so the extra upward force on the block is $F_{oil} = PA = \rho_{oil} g h A = \rho_{oil} g \Delta V$, where $\Delta V = hA$ is the volume of the block below the top surface of the oil.

The force from the oil and the buoyant force of water balance the tension and the weight of the block:

$$\sum F_y = 0: \quad F_{oil} + B - T - F_g = 0$$
$$F_{oil} + 25.0\ \text{N} - 60.0\ \text{N} - 15.0\ \text{N} = 0$$
$$F_{oil} = 50.0\ \text{N}$$

The ratio of F_{oil} and B are

$$\frac{F_{up}}{B} = \frac{\rho_{oil} g \Delta V}{\rho_{water} g (V/4)} \rightarrow \frac{\Delta V}{V} = \frac{F_{up}}{4B} \frac{\rho_{water}}{\rho_{oil}}$$

$$\frac{\Delta V}{V} = \frac{50.0\ \text{N}}{4(25.0\ \text{N})} \frac{1\,000\ \text{kg/m}^3}{800\ \text{kg/m}^3} = 0.625$$

The additional fraction of the block's volume below the top surface of the oil is $\boxed{62.5\%.}$

P14.35 (a) Since the balloon is fully submerged in air, $V_{submerged} = V_b = 325\ \text{m}^3$, and

$$B = \rho_{air} g V_b = (1.20\ \text{kg/m}^3)(9.80\ \text{m/s}^2)(325\ \text{m}^3)$$
$$= \boxed{3.82 \times 10^3\ \text{N}}$$

(b) $$\sum F_y = B - w_b - w_{He} = B - m_b g - \rho_{He} g V_b = B - (m_b + \rho_{He} V) g$$
$$= 3.82 \times 10^3\ \text{N} - \left[226\ \text{kg} + (0.179\ \text{kg/m}^3)(325\ \text{m}^3)\right](9.80\ \text{m/s}^2)$$
$$= \boxed{+1.04 \times 10^3\ \text{N}}$$

Since $\sum F_y = m a_y > 0$, a_y will be positive (upward), and $\boxed{\text{the balloon rises}}$.

(c) If the balloon and load are in equilibrium,

$$\sum F_y = (B - w_b - w_{He}) - w_{load} = 0$$

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and

$$w_{\text{load}} = \left(B - w_b - w_{\text{He}}\right) = 1.04 \times 10^3 \text{ N}$$

Thus, the mass of the load is

$$m_{\text{load}} = \frac{w_{\text{load}}}{g} = \frac{1.04 \times 10^3 \text{ N}}{9.80 \text{ m/s}^2} = \boxed{106 \text{ kg}}$$

P14.36 Let A represent the horizontal cross-sectional area of the rod, which we presume to be constant. The rod is in equilibrium:

$$\sum F_y = 0: \quad -mg + B = 0 = -\rho_0 V_{\text{whole rod}} g + \rho_{\text{fluid}} V_{\text{immersed}} g$$

$$\rho_0 ALg = \rho A\left(L - h\right)g$$

The density of the liquid is $\rho = \dfrac{\rho_0 L}{L - h}$.

P14.37 We use the result of Problem 14.36. For the rod floating in a liquid of density 0.98 g/cm^3,

$$\rho = \rho_0 \frac{L}{L - h}$$

$$0.98 \text{ g/cm}^3 = \frac{\rho_0 L}{(L - 0.2 \text{ cm})}$$

$$\left(0.98 \text{ g/cm}^3\right)L - \left(0.98 \text{ g/cm}^3\right)0.2 \text{ cm} = \rho_0 L$$

For floating in the dense liquid,

$$1.14 \text{ g/cm}^3 = \frac{\rho_0 L}{(L - 1.80 \text{ cm})}$$

$$\left(1.14 \text{ g/cm}^3\right)L - \left(1.14 \text{ g/cm}^3\right)(1.80 \text{ cm}) = \rho_0 L$$

(a) By substitution, and suppressing units,

$$1.14L - 1.14(1.80) = 0.98L - 0.200(0.98)$$

$$0.16L = 1.856$$

$$L = \boxed{11.6 \text{ cm}}$$

(b) Substituting back,

$$\left(0.98 \text{ g/cm}^3\right)(11.6 \text{ cm} - 0.200 \text{ cm}) = \rho_0 (11.6 \text{ cm})$$

$$\rho_0 = \boxed{0.963 \text{ g/cm}^3}$$

(c) $\boxed{\text{No; the density } \rho \text{ is not linear in } h.}$

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P14.38 (a) We can estimate the total buoyant force of the 600 toy balloons as

$$B_{\text{total}} = 600 \cdot B_{\substack{\text{single}\\ \text{balloon}}} = 600\left(\rho_{\text{air}} g V_{\text{balloon}}\right)$$

$$= 600\left[\rho_{\text{air}} g\left(\frac{4\pi}{3} r^3\right)\right]$$

$$= 600\left[\left(1.20\ \text{kg/m}^3\right)\left(9.80\ \text{m/s}^2\right)\frac{4\pi}{3}(0.50\ \text{m})^3\right]$$

$$= 3.7 \times 10^3\ \text{N} = \boxed{3.7\ \text{kN}}$$

(b) We estimate the net upward force by applying Newton's second law in the vertical direction:

$$\sum F_y = B_{\text{total}} - m_{\text{total}} g$$

$$= 3.7 \times 10^3\ \text{N} - 600\left(0.30\ \text{kg}\right)\left(9.8\ \text{m/s}^2\right)$$

$$= 1.9 \times 10^3\ \text{N} = \boxed{1.9\ \text{kN}}$$

This net force was sufficient to lift Ashpole, his parachute, and other supplies.

(c) Atmospheric pressure at this high altitude is much lower than at Earth's surface, so the balloons expanded and eventually burst.

P14.39 We assume that the mass of the balloon envelope is included in the 400 kg. We assume that the 400-kg total load is much denser than air and so has negligible volume compared to the helium. At z = 8 000 m, the density of air is

$$\rho_{\text{air}} = \rho_0 e^{-z/8\ 000} = (1.20\ \text{kg/m}^3)e^{-1}$$

$$= (1.20\ \text{kg/m}^3)(0.368)$$

$$= 0.441\ \text{kg/m}^3$$

Think of the balloon reaching equilibrium at this height. The weight of its payload is Mg = (400 kg)(9.80 m/s^2) = 3 920 N. The weight of the helium in it is $mg = \rho_{\text{He}} V g$.

$$\sum F_y = 0 \quad \rightarrow \quad +\rho_{\text{air}} V g - Mg - \rho_{\text{He}} V g = 0$$

Solving,

$$(\rho_{\text{air}} - \rho_{\text{He}})V = M$$

and

$$V = \frac{M}{\rho_{\text{air}} - \rho_{\text{He}}} = \frac{400\ \text{kg}}{(0.441 - 0.179)\ \text{kg/m}^3} = \boxed{1.52 \times 10^3\ \text{m}^3}$$

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Section 14.5 Fluid Dynamics
Section 14.6 Bernoulli's Equation

P14.40 (a) The cross-sectional area of the hose is

$$A = \pi r^2 = \pi d^2 / 4 = \pi (2.74 \text{ cm})^2 / 4$$

and the volume flow rate (volume per unit time) is

$$Av = 25.0 \text{ L}/1.50 \text{ min}$$

Thus,

$$v = \frac{25.0 \text{ L}/1.50 \text{ min}}{A}$$

$$= \left(\frac{25.0 \text{ L}}{1.50 \text{ min}}\right)\left[\frac{4}{\pi \cdot (2.74)^2 \text{ cm}^2}\right]\left(\frac{1 \text{ min}}{60 \text{ s}}\right)\left(\frac{10^3 \text{ cm}^3}{1 \text{ L}}\right)$$

$$= (47.1 \text{ cm/s})\left(\frac{1 \text{ m}}{10^2 \text{ cm}}\right) = \boxed{0.471 \text{ m/s}}$$

(b) $$\frac{A_2}{A_1} = \left(\frac{\pi d_2^2}{4}\right)\left(\frac{4}{\pi d_1^2}\right) = \left(\frac{d_2}{d_1}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9} \quad \text{or} \quad A_2 = \frac{A_1}{9}$$

Then from the equation of continuity, $A_2 v_2 = A_1 v_1$, we find

$$v_2 = \left(\frac{A_1}{A_2}\right) v_1 = 9(0.471 \text{ m/s}) = \boxed{4.24 \text{ m/s}}$$

P14.41 Assuming the top is open to the atmosphere, then

$$P_1 = P_0$$

Note $P_2 = P_0$. The water pushes on the air just as hard as the air pushes on the water.

$$\text{Flow rate} = 2.50 \times 10^{-3} \text{ m}^3/\text{min} = 4.17 \times 10^{-5} \text{ m}^3/\text{s}$$

(a) $A_1 >> A_2$ so $v_1 << v_2$

Assuming $v_1 = 0$,

$$P_1 + \frac{\rho v_1^2}{2} + \rho g y_1 = P_2 + \frac{\rho v_2^2}{2} + \rho g y_2$$

$$v_2 = \sqrt{2 g y_1} = \sqrt{2(9.80 \text{ m/s}^2)(16.0 \text{ m})} = \boxed{17.7 \text{ m/s}}$$

(b) $$\text{Flow rate} = A_2 v_2 = \left(\frac{\pi d^2}{4}\right)(17.7 \text{ m/s}) = 4.17 \times 10^{-5} \text{ m}^3/\text{s}$$

$$d = \boxed{1.73 \times 10^{-3} \text{ m}} = 1.73 \text{ mm}$$

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P14.42 (a) The mass flow rate and the volume flow rate are constant:

$$\rho A_1 v_1 = \rho A_2 v_2 \quad \rightarrow \quad \pi r_1^2 v_1 = \pi r_2^2 v_2$$

Substituting,

$$(3.00\text{ cm})^2 v_1 = (1.50\text{ cm})^2 v_2 \quad \rightarrow \quad v_2 = 4v_1$$

For ideal flow,

$$P_1 + \rho g y_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2}\rho v_2^2$$

$$1.75\times 10^4\text{ Pa} + 0 + \frac{1}{2}\left(1\,000\text{ kg/m}^3\right)\left(v_1\right)^2 = 1.20\times 10^4\text{ Pa} + (1000)(9.8)(0.250)\text{ Pa} + \frac{1}{2}\left(1\,000\text{ kg/m}^3\right)(4v_1)^2$$

Solving for v_1 gives

$$v_1 = \sqrt{\frac{3\,050\text{ Pa}}{7\,500\text{ kg/m}^3}} = \boxed{0.638\text{ m/s}}$$

(b) From part (a), we have

$$v_2 = 4v_1 = \boxed{2.55\text{ m/s}}$$

(c) The volume flow rate is

$$\pi r_1^2 v_1 = \pi(0.030\,0\text{ m})^2(0.638\text{ m/s}) = \boxed{1.80\times 10^{-3}\text{ m}^3/\text{s}}$$

P14.43 The volume flow rate is

$$\frac{\Delta V}{\Delta t} = \frac{125\text{ cm}^3}{16.3\text{ s}} = 7.67\text{ cm}^3/\text{s} = Av_1$$

where $d = 0.96$ cm and $A = \pi r^2 = 0.724\text{ cm}^2$. The speed at the top of the falling column is

$$v_1 = \frac{\Delta V/\Delta t}{A} = \frac{7.67\text{ cm}^3/\text{s}}{0.724\text{ cm}^3} = 10.6\text{ cm/s}$$

Take point 2 at 13 cm below:

$$P_1 + \rho g y_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2}\rho v_2^2$$

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$$P_0 + (1\,000\ \text{kg/m}^3)(9.80\ \text{m/s}^2)0.130\ \text{m} + \frac{1}{2}(1\,000\ \text{kg/m}^3)(0.106\ \text{m/s})^2 = P_0 + 0 + \frac{1}{2}(1\,000\ \text{kg/m}^3)v_2^2$$

solving for the velocity gives

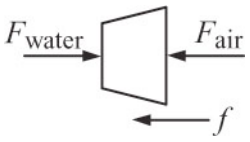
$$v_2 = \sqrt{2(9.80\ \text{m/s}^2)(0.130\ \text{m}) + (0.106\ \text{m/s})^2} = 1.60\ \text{m/s}$$

The volume flow rate is constant:

$$7.67\ \text{cm}^3/\text{s} = \pi\left(\frac{d}{2}\right)^2 160\ \text{cm/s}$$

$$d = \boxed{0.247\ \text{cm}}$$

P14.44 Take point ① at the free surface of the water in the tank and ② inside the nozzle.



ANS. FIG. P14.44

(a) With the cork in place,

$$P_1 + \rho g y_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2}\rho v_2^2$$

becomes

$$P_0 + (1\,000\ \text{kg/m}^3)(9.80\ \text{m/s}^2)(7.50\ \text{m}) + 0 = P_2 + 0 + 0$$

$$P_2 - P_0 = 7.35 \times 10^4\ \text{Pa}$$

For the stopper,

$$\sum F_x = 0$$

$$F_{\text{water}} - F_{\text{air}} - f = 0$$

$$P_2 A - P_0 A = f$$

$$f = (7.35 \times 10^4\ \text{Pa})\pi(0.011\,0\ \text{m})^2 = \boxed{27.9\ \text{N}}$$

(b) Now Bernoulli's equation gives

$$P_0 + 7.35 \times 10^4\ \text{Pa} + 0 = P_0 + 0 + \frac{1}{2}(1\,000\ \text{kg/m}^3)v_2^2$$

$$v_2 = 12.1\ \text{m/s}$$

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The quantity leaving the nozzle in 2 h is

$$\rho V = \rho A v_2 t$$

$$= \left(1\ 000\ \text{kg/m}^3\right)\pi(0.011\ 0\ \text{m})^2(12.1\ \text{m/s})(7\ 200\ \text{s})$$

$$= \boxed{3.32\times10^4\ \text{kg}}$$

(c) Take point 1 in the wide hose and 2 just outside the nozzle. Applying the continuity equation:

$$A_1 v_1 = A_2 v_2$$

$$\pi\left(\frac{6.60\ \text{cm}}{2}\right)^2 v_1 = \pi\left(\frac{2.20\ \text{cm}}{2}\right)^2 (12.1\ \text{m/s})$$

$$v_1 = \frac{12.1\ \text{m/s}}{9} = 1.35\ \text{m/s}$$

$$P_1 + \rho g y_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2}\rho v_2^2$$

$$P_1 + 0 + \frac{1}{2}\left(1\ 000\ \text{kg/m}^3\right)(1.35\ \text{m/s})^2$$

$$= P_0 + 0 + \frac{1}{2}\left(1\ 000\ \text{kg/m}^3\right)(12.1\ \text{m/s})^2$$

$$P_1 - P_0 = 7.35\times10^4\ \text{Pa} - 9.07\times10^2\ \text{Pa} = \boxed{7.26\times10^4\ \text{Pa}}$$

P14.45 (a) Between sea surface and clogged hole:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

$$1\ \text{atm} + 0 + \left(1\ 030\ \text{kg/m}^3\right)\left(9.80\ \text{m/s}^2\right)(2.00\ \text{m}) = P_2 + 0 + 0$$

$$P_2 = 1\ \text{atm} + 20.2\ \text{kPa}$$

The air on the back of his hand pushes opposite the water, so the net force on his hand is

$$F = PA = \left(20.2\times10^3\ \text{N/m}^2\right)\left(\frac{\pi}{4}\right)\left(1.2\times10^{-2}\ \text{m}\right)^2$$

$$F = \boxed{2.28\ \text{N}}\ \text{toward Holland}$$

(b) Now, Bernoulli's equation gives

$$1\ \text{atm} + 0 + 20.2\ \text{kPa} = 1\ \text{atm} + \frac{1}{2}\left(1\ 030\ \text{kg/m}^3\right)v_2^2 + 0$$

$$v_2 = 6.26\ \text{m/s}$$

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The volume rate of flow is

$$A_2 v_2 = \frac{\pi}{4}\left(1.2\times10^{-2}\text{ m}\right)^2(6.26\text{ m/s}) = 7.08\times10^{-4}\text{ m}^3/\text{s}$$

One acre–foot is $4\ 047\text{ m}^2 \times 0.304\ 8\text{ m} = 1\ 234\text{ m}^3$.

Requiring $\dfrac{1\ 234\text{ m}^3}{7.08\times10^{-4}\text{ m}^3/\text{s}} = \boxed{1.74\times10^6\text{ s}} = 20.2\text{ days}.$

P14.46 (a) Power is the rate of energy flow as a function of time:

$$P = \frac{\Delta E}{\Delta t} = \frac{\Delta mgh}{\Delta t} = \left(\frac{\Delta m}{\Delta t}\right)gh = Rgh$$

(b) The power delivered by the Grand Coulee dam is

$$P_{\text{EL}} = 0.85\left(8.50\times10^5\text{ kg/s}\right)\left(9.80\text{ m/s}^2\right)(87.0\text{ m}) = \boxed{616\text{ MW}}$$

P14.47 (a) The cross-sectinal area is the same everywhere, so the speed is the same everywhere:

$$\left(P+\frac{1}{2}\rho v^2+\rho gy\right)_{\text{river}} = \left(P+\frac{1}{2}\rho v^2+\rho gy\right)_{\text{rim}}$$

$$P + 0 + \rho g(564\text{ m}) = 1\text{ atm} + 0 + \rho g(2\ 096\text{ m})$$

$$P = 1\text{ atm} + \left(1\ 000\text{ kg/m}^3\right)\left(9.80\text{ m/s}^2\right)(1\ 532\text{ m})$$

$$= \boxed{1\text{ atm} + 15.0\text{ MPa}}$$

(b) The volume flow rate is $4\ 500\text{ m}^3/d = Av = \dfrac{\pi d^2 v}{4}$.

$$v = \left(4\ 500\text{ m}^3/\text{d}\right)\left(\frac{1\text{ d}}{86\ 400\text{ s}}\right)\left(\frac{4}{\pi(0.150\text{ m})^2}\right) = \boxed{2.95\text{ m/s}}$$

P14.48 (a) The volume flow rate is the same at the two points: $A_1 v_1 = A_2 v_2$:

$$\pi(1\text{ cm})^2 v_1 = \pi(0.5\text{ cm})^2 v_2 \quad\rightarrow\quad v_2 = 4v_1$$

We assume the tubes are at the same elevation:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

$$P_1 - P_2 = \Delta P = \frac{1}{2}\rho(4v_1)^2 + 0 - \frac{1}{2}\rho v_1^2$$

$$\Delta P = \frac{1}{2}(850\text{ kg/m}^3)\ 15v_1^2$$

$$v_1 = (0.012\ 5\text{ m/s})\sqrt{\Delta P}$$

where the pressure is in pascals.

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The volume flow rate is

$$\pi(0.01\text{ m})^2(0.0125\text{ m/s})\sqrt{\Delta P}$$

$$= \boxed{\left(3.93\times10^{-6}\text{ m}^3/\text{s}\right)\sqrt{\Delta P},\quad \text{where } \Delta P \text{ is in pascals}}$$

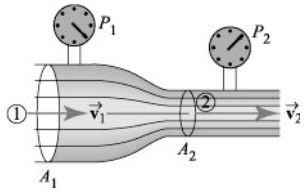
(b) For $\Delta P = 6.00$ kPa,

$$\left(3.93\times10^{-6}\text{ m}^3/\text{s}\right)\sqrt{6\ 000\text{ Pa}} = \boxed{0.305\text{ L/s}}$$

(c) With pressure difference 2 times larger, the flow rate is larger by the square root of 2:

$$\sqrt{2}(0.305\text{ L/s}) = \boxed{0.431\text{ L/s}}$$

P14.49 (a) Since the tube is horizontal, $y_1 = y_2$ and the gravity terms in Bernoulli's equation cancel, leaving



ANS. FIG. P14.49

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

or

$$v_2^2 - v_1^2 = \frac{2(P_1 - P_2)}{\rho} = \frac{2\left(1.20\times10^3\text{ Pa}\right)}{7.00\times10^2\text{ kg/m}^3}$$

and

$$v_2^2 - v_1^2 = 3.43\text{ m}^2/\text{s}^2 \qquad \textbf{[1]}$$

From the continuity equation, $A_1 v_1 = A_2 v_2$, we find

$$v_2 = \left(\frac{A_1}{A_2}\right)v_1 = \left(\frac{r_1}{r_2}\right)^2 v_1 = \left(\frac{2.40\text{ cm}}{1.20\text{ cm}}\right)^2 v_1$$

or

$$v_2 = 4v_1 \qquad \textbf{[2]}$$

Substituting equation [2] into [1] yields $15v_1^2 = 3.43\text{ m}^2/\text{s}^2$ and

$$v_1 = 0.478\text{ m/s}$$

Then, equation [2] gives

$$v_2 = 4(0.478\text{ m/s}) = \boxed{1.91\text{ m/s}}$$

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(b) The volume flow rate is

$$A_1 v_1 = A_2 v_2 = \left(\pi r_2^2\right) v_2 = \pi\left(1.20\times10^{-2}\ \text{m}\right)^2 (1.91\ \text{m/s})$$
$$= \boxed{8.64\times10^{-4}\ \text{m}^3/\text{s}}$$

P14.50 (a) For upward flight of a water-drop projectile from geyser vent to fountain–top, $v_{yf}^2 = v_{yi}^2 + 2a_y\Delta y$.

Then $0 = v_i^2 + 2\left(-9.80\ \text{m/s}^2\right)(+40.0\ \text{m})$ and $v_i = \boxed{28.0\ \text{m/s}}$.

(b) Between geyser vent and fountain-top:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

Air is so low in density that very nearly $P_1 = P_2 = 1$ atm. Then,

$$\frac{1}{2}v_1^2 + 0 = 0 + \left(9.80\ \text{m/s}^2\right)(40.0\ \text{m})$$

$$v_1 = \boxed{28.0\ \text{m/s}}$$

(c) The answers agree precisely. The models are consistent with each other.

(d) Between the chamber and the fountain-top:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

$$P_1 + 0 + \left(1\,000\ \text{kg/m}^3\right)\left(9.80\ \text{m/s}^2\right)(-175\ \text{m})$$
$$= P_0 + 0 + \left(1\,000\ \text{kg/m}^3\right)\left(9.80\ \text{m/s}^2\right)(+40.0\ \text{m})$$
$$P_1 - P_0 = \left(1\,000\ \text{kg/m}^3\right)\left(9.80\ \text{m/s}^2\right)(215\ \text{m}) = \boxed{2.11\ \text{MPa}}$$

Section 14.7 Other Applications of Fluid Dynamics

P14.51 The assumption of incompressibility is surely unrealistic, but allows an estimate of the speed. From Bernoulli's equation,

$$P_1 + \rho g y_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2}\rho v_2^2$$

$$1.00\ \text{atm} + 0 + 0 = 0.287\ \text{atm} + 0 + \frac{1}{2}\left(1.20\ \text{kg/m}^3\right)v_2^2$$

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solving for the velocity gives

$$v_2 = \sqrt{\frac{2(1.00 - 0.287)\left(1.013 \times 10^5 \text{ N/m}^2\right)}{1.20 \text{ kg/m}^3}} = \boxed{347 \text{ m/s}}$$

P14.52 (a) Force balance requires that

$$Mg = \left(P_1 - P_2\right)A$$

$$\frac{\left(16\,000 \text{ kg}\right)\left(9.80 \text{ m/s}^2\right)}{2\left(40.0 \text{ m}^2\right)} = 7.00 \times 10^4 \text{ Pa} - P_2$$

$$\therefore P_2 = 7.0 \times 10^4 \text{ Pa} - 0.196 \times 10^4 \text{ Pa} = \boxed{6.80 \times 10^4 \text{ Pa}}$$

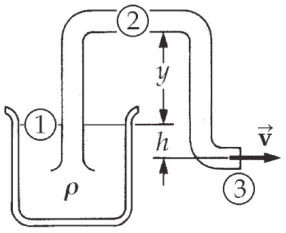
(b) Higher. With the inclusion of another upward force due to deflection of air downward, the pressure difference does not need to be as great to keep the airplane in flight.

P14.53 (a) We use Bernoulli's equation,

$$P_0 + \rho g h + 0 = P_0 + 0 + \frac{1}{2}\rho v_3^2$$

which gives $v_3 = \sqrt{2gh}$.

If $h = 1.00$ m, then $v_3 = \boxed{4.43 \text{ m/s}}$.



ANS. FIG. P14.53

(b) Again, from Bernoulli's equation,

$$P + \rho g y + \frac{1}{2}\rho v_2^2 = P_0 + 0 + \frac{1}{2}\rho v_3^2$$

Since $v_2 = v_3$,

$$P = P_0 - \rho g y$$

Since $P \geq 2.3$ kPa, the greatest possible siphon height is given by

$$y \leq \frac{P_0 - P}{\rho g} = \frac{1.013 \times 10^5 \text{ Pa} - 2.30 \times 10^3 \text{ Pa}}{\left(10^3 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)} = \boxed{10.1 \text{ m}}$$

P14.54 Take points 1 and 2 in the air just inside and outside the window pane.

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

$$P_0 + 0 = P_2 + \frac{1}{2}\left(1.20 \text{ kg/m}^3\right)\left(11.2 \text{ m/s}\right)^2 \quad \rightarrow \quad P_2 = P_0 - 75.3 \text{ Pa}$$

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(a) The total force exerted by the air is outward,

$$P_1A - P_2A = P_0A - P_0A + (75.3\ \text{N/m}^2)(4.00\ \text{m})(1.50\ \text{m})$$

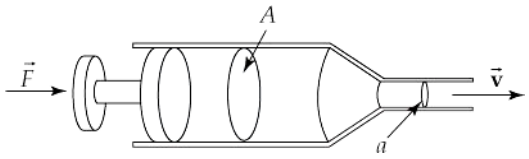
$$= \boxed{452\ \text{N outward}}$$

(b) $$P_1A - P_2A = \frac{1}{2}\rho v_2^2 A = \frac{1}{2}(1.20\ \text{kg/m}^3)(22.4\ \text{m/s})^2(4.00\ \text{m})(1.50\ \text{m})$$

$$= \boxed{1.81\ \text{kN outward}}$$

P14.55 In the reservoir, the gauge pressure is

$$\Delta P = \frac{2.00\ \text{N}}{2.50\times10^{-5}\ \text{m}^2} = 8.00\times10^4\ \text{Pa}$$



ANS. FIG. P14.55

From the equation of continuity, we have

$$A_1v_1 = A_2v_2$$

$$(2.50\times10^{-5}\ \text{m}^2)v_1 = (1.00\times10^{-8}\ \text{m}^2)v_2 \ \text{ so } \ v_1 = (4.00\times10^{-4})v_2$$

Thus, v_1^2 is negligible in comparison to v_2^2. In Bernoulli's equation, $(P_1 - P_2) + \frac{1}{2}\rho v_1^2 + \rho g y_1 = \frac{1}{2}\rho v_2^2 + \rho g y_2$, the term in v_1^2 is essentially zero and the terms in y_1 and y_2 cancel each other. Then,

$$v_2 = \left(\frac{2(P_1 - P_2)}{\rho}\right)^{1/2} = \sqrt{\frac{2(8.00\times10^4\ \text{Pa})}{1\,000\ \text{kg/m}^3}} = \boxed{12.6\ \text{m/s}}$$

Additional Problems

***P14.56** The water exerts a buoyant force on the air, given by

$$B = \rho_{\text{fluid}} g V = (1\,000\ \text{kg/m}^3)(9.80\ \text{m/s}^2)(10.0\ \text{L})\left(\frac{1\ \text{m}^3}{10^3\ \text{L}}\right)$$

$$= 98.0\ \text{N up}$$

The weight of the air is

$$F_g = \rho g V = (2.40\ \text{kg/m}^3)(9.80\ \text{m/s}^2)(10.0\times10^{-3}\ \text{m}^3)$$

$$= 0.235\ \text{N down}$$

To transport the air down at constant speed requires a downward force D in $+98.0\ \text{N} - 0.235\ \text{N} - D = 0$, $D = 97.8\ \text{N}$, and work

$$W = \vec{\mathbf{D}} \cdot \vec{\mathbf{d}} = (97.8\ \text{N})(10.3\ \text{m})\cos 0° = \boxed{1.01\ \text{kJ}}$$

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P14.57 (a) At a depth of 1 000 m,

$$P = P_0 + \rho g h = 1.013 \times 10^5 \text{ Pa} + \left(1\,030 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)\left(1\,000 \text{ m}\right)$$

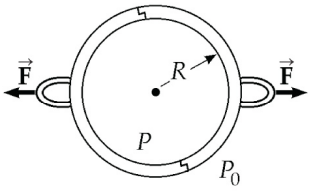
$$P = \boxed{1.02 \times 10^7 \text{ Pa}}$$

(b) The buoyant force on the submarine at this depth is

$$B = \rho g V = \rho g \frac{4}{3}\pi r^3 = \left(1\,030 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)\frac{4}{3}\pi(2.50 \text{ m})^3$$

$$= \boxed{6.61 \times 10^5 \text{ N}}$$

P14.58 The pressure on the surface of the two hemispheres is constant at all points, and the force on each element of surface area is directed along the radius of the hemispheres. The applied force along the axis must balance the force on the "effective" area, which is the projection of the actual surface onto a plane perpendicular to the x axis, $A = \pi R^2$. Therefore,



ANS. FIG. P14.58

$$F = \boxed{\left(P_0 - P\right)\pi R^2}$$

P14.59 (a) The weight of the ball must be equal to the buoyant force of the water:

$$1.26 \text{ kg } g = \rho_{\text{water}} \frac{4}{3}\pi r_{\text{outer}}^3 g$$

$$r_{\text{outer}} = \left(\frac{3 \times 1.26 \text{ kg}}{4\pi \ 1\,000 \text{ kg/m}^3} \right)^{1/3} = \boxed{6.70 \text{ cm}}$$

(b) The mass of the ball is determined by the density of aluminum:

$$m = \rho_{\text{Al}} V = \rho_{\text{Al}} \left(\frac{4}{3}\pi r_0^3 - \frac{4}{3}\pi r_i^3 \right)$$

$$1.26 \text{ kg} = 2700 \text{ kg/m}^3 \left(\frac{4}{3}\pi \right)\left((0.067 \text{ m})^3 - r_i^3 \right)$$

$$1.11 \times 10^{-4} \text{ m}^3 = 3.01 \times 10^{-4} \text{ m}^3 - r_i^3$$

$$r_i = \left(1.89 \times 10^{-4} \text{ m}^3\right)^{1/3} = \boxed{5.74 \text{ cm}}$$

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P14.60 (a) A particle in equilibrium model

(b) When the balloon comes into equilibrium, we must have

$$\boxed{\sum F_y = B - F_b - F_{\text{He}} - F_s = 0}$$

where B is the buoyant force, F_b the weight of the balloon, F_{He} the weight of the helium, and F_s the weight of the segment of string above the ground.

(c) Write expressions for each of the terms in the force equation:

$$B = \rho_{\text{air}} V g = \rho_{\text{air}} \frac{4}{3}\pi r^3 g$$

$$F_b = m_b g$$

$$F_{\text{He}} = \rho_{\text{He}} V g = \rho_{\text{He}} \frac{4}{3}\pi r^3 g$$

and $F_s = m_s g$; where $m_s = m\dfrac{h}{\ell}$

Therefore, we have

$$\rho_{\text{air}} V \cancel{g} - m_b \cancel{g} - \rho_{\text{He}} V \cancel{g} - m_s \cancel{g} = 0$$

or $$m_s = \left(\rho_{\text{air}} - \rho_{\text{He}}\right) V - m_b \rightarrow \boxed{m_s = \left(\rho_{\text{air}} - \rho_{\text{He}}\right)\frac{4}{3}\pi r^2 - m_b}$$

(d) $$m_s = \left[(1.20 - 0.179)\ \text{kg/m}^3\right]\left[\frac{4}{3}\pi (0.400\ \text{m})^3\right] - 0.250\ \text{kg}$$

$$= \boxed{0.023\ 7\ \text{kg}}$$

(e) $$m_s = m\frac{h}{\ell} \rightarrow h = \ell\frac{m_s}{m} = (2.00\ \text{m})\ \frac{0.023\ 7\ \text{kg}}{0.050\ 0\ \text{kg}} = \boxed{0.948\ \text{m}}$$

P14.61 Consider the diagram in ANS. FIG. P14.61 and apply Bernoulli's equation to points A and B, taking $y = 0$ at the level of point B, and recognizing that v_A is approximately zero. This gives:

$$P_A + \frac{1}{2}\rho_w (0)^2 + \rho_w g (h - L\sin\theta) = P_B + \frac{1}{2}\rho_w v_B^2 + \rho_w g (0)$$

Now, recognize that $P_A = P_B = P_{\text{atmosphere}}$ since both points are open to the atmosphere (neglecting variation of atmospheric pressure with altitude). Thus, we obtain

$$v_B = \sqrt{2g(h - L\sin\theta)}$$

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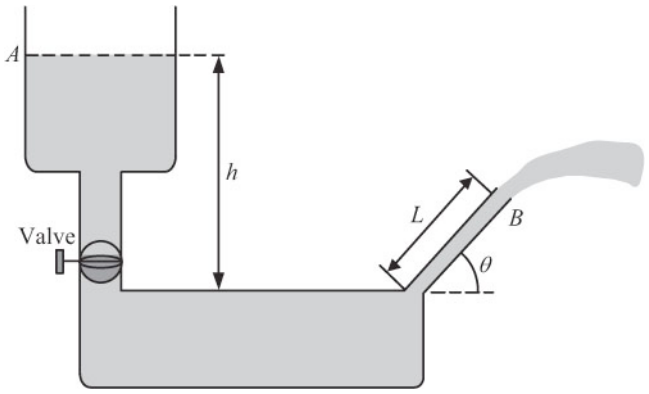
Now the problem reduces to one of projectile motion with $v_{yi} = v_B \sin\theta$. Then using, $v_{yf}^2 = v_{yi}^2 + 2a(\Delta y)$, where $y = y_{max}, v_{yf} = 0,$ and $a = -g$, we find

$$\Delta y = \frac{0 - v_{yi}^{\,2}}{2a} = \frac{-v_B^{\,2}\sin^2\theta}{2(-g)} = \frac{[2g(h - L\sin\theta)]\sin^2\theta}{2g}$$

$$\Delta y = (h - L\sin\theta)\sin^2\theta$$

$$\Delta y = [10.0\text{ m} - (2.00\text{ m})\sin 30.0°]\sin^2 30.0°$$

$$y_{max} = \boxed{2.25\text{ m (above the level where the water emerges)}}$$



ANS. FIG. P14.61

P14.62 The "balanced" condition is one in which the apparent weight of the body equals the apparent weight of the weights. This condition can be written as

$$F_g - B = F_g' - B'$$

where B and B' are the buoyant forces on the body and weights, respectively. The buoyant force experienced by an object of volume V in air equals

$$\text{Buoyant force} = (\text{Volume of object})\rho_{air} g$$

so we have $B = V\rho_{air} g$ and

$$B' = \left(\frac{F_g'}{\rho g}\right)\rho_{air} g$$

Therefore, $F_g = F_g' + \left(V - \frac{F_g'}{\rho g}\right)\rho_{air} g.$

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P14.63 Assume $v_{\text{inside}} \approx 0$. From Bernoulli's equation,

$$P + 0 + 0 = 1 \text{ atm} + \frac{1}{2}\left(1\,000 \text{ kg/m}^3\right)\left(30.0 \text{ m/s}\right)^2 + \left(1\,000 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)\left(0.500 \text{ m}\right)$$

$$P_{\text{gauge}} = P - 1 \text{ atm} = 4.50 \times 10^5 \text{ Pa} + 4.90 \times 10^3 \text{ Pa}$$

$$= \boxed{455 \text{ kPa}}$$

P14.64 Let the ball be released at point 1, enter the liquid at point 2, attain maximum depth at point 3, and pop through the surface on the way up at point 4.

(a) Energy conservation for the fall through the air:

$$K_i + U_i = K_f + U_f$$

$$0 + mgy_1 = \frac{1}{2}mv_2^2$$

$$v_2 = \sqrt{2gy_1} = \sqrt{2\left(9.80 \text{ m/s}^2\right)(3.30 \text{ m})} = \boxed{8.04 \text{ m/s}}$$

(b) The gravitational force and the buoyant force.

The gravitational force is

$$mg = (2.10 \text{ kg})(9.80 \text{ N/kg}) = 20.6 \text{ N down}$$

and the buoyant force is

$$m_{\text{fluid}}g = \rho_{\text{fluid}}V_{\text{object}}g = \rho_{\text{fluid}}(4/3)\pi r^3 g$$

$$= \left(1\,230 \text{ kg/m}^3\right)(4\pi/3)(0.090\,0 \text{ m})^3\left(9.80 \text{ m/s}^2\right)$$

$$= 36.8 \text{ N up}$$

(c) The buoyant force is greater than the gravitational force.

The net upward force on the ball brings its downward motion to a stop.

We choose to use the work-kinetic energy theorem.

$$\frac{1}{2}mv_2^2 + F_{net} \cdot \Delta y = \frac{1}{2}mv_3^2$$

$$\frac{1}{2}\left(2.10 \text{ kg}\right)\left(8.04 \text{ m/s}\right)^2 + (36.8 \text{ N} - 20.6 \text{ N})\left(-\Delta y\right) = 0$$

$$\Delta y = 67.9 \text{ J}/16.2 \text{ N} = \boxed{4.18 \text{ m}}$$

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(d) The same net force acts on the ball over the same distance as it moves down and as it moves up, to produce the same speed change. Thus $v_4 =$ $\boxed{8.04 \text{ m/s}}$.

(e) $\boxed{\text{The time intervals are equal}}$, because the ball moves with the same range of speeds over equal distance intervals.

(f) $\boxed{\text{With friction present, } \Delta t_{\text{down}} \text{ is less than } \Delta t_{\text{up}}. \text{ The magnitude of the ball's acceleration on the way down is greater than its acceleration on the way up. The two motions cover equal distances and both have zero speed at one end point, so the downward trip with larger-magnitude acceleration must take less time.}}$

P14.65 At equilibrium, $\sum F_y = 0$: $\quad B - F_{\text{spring}} - F_{g,\,\text{He}} - F_{g,\,\text{balloon}} = 0$

giving $\quad F_{\text{spring}} = kL = B - \left(m_{\text{He}} + m_{\text{balloon}}\right) g$

But B = weight of displaced air = $\rho_{\text{air}} V g$

and $\quad m_{\text{He}} = \rho_{\text{He}} V$

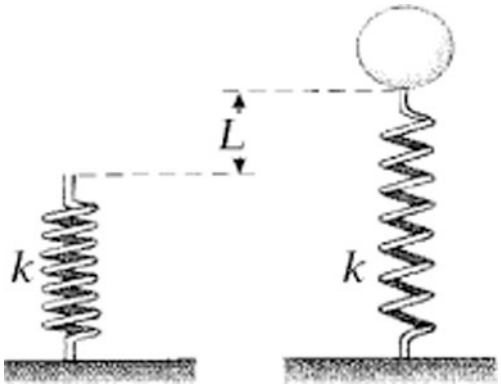
Therefore, we have $kL = \rho_{\text{air}} V g - \rho_{\text{He}} V g - m_{\text{balloon}} g$

or $\quad L = \dfrac{\left(\rho_{\text{air}} - \rho_{\text{He}}\right) V - m_{\text{balloon}}}{k} g$

From the data given,

$$L = \frac{\left[(1.20 - 0.179)\ \text{kg/m}^3\right]\left(5.00\ \text{m}^3\right) - 2.00 \times 10^{-3}\ \text{kg}}{90.0\ \text{N/m}} \left(9.80\ \text{m/s}^2\right)$$

Thus, this gives $\boxed{L = 0.556 \text{ m.}}$



ANS. FIG. P14.65

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***P14.66** Consider spherical balloons of radius 12.5 cm containing helium at STP and immersed in air at 0°C and 1 atm. If the rubber envelope has mass 5.00 g, the upward force on each is

$$F_{up} = B - F_{g,\text{He}} - F_{g,\text{env}} = \rho_{\text{air}} V g - \rho_{\text{He}} V g - m_{\text{env}} g$$

$$F_{up} = (\rho_{\text{air}} - \rho_{\text{He}})\left(\frac{4}{3}\pi r^3\right) g - m_{\text{env}} g$$

$$F_{up} = \left[(1.29 - 0.179)\ \text{kg/m}^3\right]\left[\frac{4}{3}\pi (0.125\ \text{m})^3\right](9.80\ \text{m/s}^2)$$
$$-(5.00\times 10^{-3}\ \text{kg})(9.80\ \text{m/s}^2) = 0.040\,1\ \text{N}$$

If your weight (including harness, strings, and submarine sandwich) is

$$70.0\ \text{kg}(9.80\ \text{m/s}^2) = 686\ \text{N}$$

you need this many balloons:

$$\frac{686\ \text{N}}{0.040\,1\ \text{N}} = 17\ 000 \boxed{\sim 10^4}$$

P14.67 The buoyant force B supports the weights of the raft and the boy. Using M = mass of boy, V = volume of raft, ρ_{st} = density of Styrofoam, and ρ_{w} = density of water, and the volume of the raft is

$$V = (1.00\ \text{m})(1.00\ \text{m})(0.050\ \text{m}) = 0.050\ \text{m}^3$$

From Newton's second law,

$$\sum F_y = 0: \quad B - Mg - \rho_{\text{st}} g V = 0 \quad \rightarrow \quad \rho_{\text{w}} g V - Mg - \rho_{\text{st}} g V = 0$$

Solving for ρ_{st} we get

$$\rho_{\text{st}} = \rho_{\text{w}} - \frac{M}{V} = 1\ 000\ \text{kg/m}^3 - \left(\frac{42.0\ \text{kg}}{0.050\ 0\ \text{m}^3}\right) = \boxed{160\ \text{kg/m}^3}$$

***P14.68** (a) The blood flowing through the artery is similar to water flowing through a pipe. We substitute numerical values into the equation for the Reynolds number:

$$\text{Re} = \frac{(1.06 \times 10^3\ \text{kg/m}^3)(6.70 \times 10^{-2}\ \text{m/s})(3.00 \times 10^{-2}\ \text{m})}{3.00 \times 10^{-3}\ \text{Pa}\cdot\text{s}}$$
$$= 710$$

Because this result is less than 2 300, the flow is laminar.

(b) Denote the situation in part (a) using subscripts 1. In the expression for the Reynolds number for the capillary, which we

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denote as situation 2, incorporate the continuity equation for fluids as the blood flows into the smaller blood vessel:

$$\mathrm{Re}_2 = \frac{\rho v_2 d_2}{\mu} = \frac{\rho v_1 \left(\frac{A_1}{A_2}\right)(2r_2)}{\mu} = \frac{\rho v_1 \left(\frac{\pi r_1^2}{\pi r_2^2}\right)(2r_2)}{\mu} = \frac{2\rho v_1 r_1^2}{\mu r_2}$$

Solve the resulting equation for the radius of the capillary:

$$r_2 = \frac{2\rho v_1 r_1^2}{\mu(\mathrm{Re}_2)}$$

Substitute numerical values, including a Reynolds number representing turbulent flow:

$$r_2 = \frac{2\left(1.06 \times 10^3 \text{ kg/m}^3\right)\left(6.70 \times 10^{-2} \text{ m/s}\right)\left(1.50 \times 10^{-2} \text{ m}\right)^2}{\left(3.00 \times 10^{-3} \text{ Pa} \cdot \text{s}\right)(4\,000)}$$

$$= \boxed{2.66 \times 10^{-3} \text{ m}}$$

(c) The situation in the human body is not represented by a large artery feeding into a single capillary as in part (b). The artery branches into smaller vessels and eventually into approximately 10 billion capillaries. Even though the radius of each capillary is very small, the overall area through which the blood flows in all the capillaries is larger than the area of the artery in part (a). Consequently, in the expression for the Reynolds number, both the speed of the blood and the diameter is very small for each capillary, representing a very low value for the Reynolds number and, consequently, laminar flow.

P14.69 (a) $P = \rho g h$ gives $1.013 \times 10^5 \text{ Pa} = \left(1.29 \text{ kg/m}^3\right)\left(9.80 \text{ m/s}^2\right)h.$

$h = \boxed{8.01 \text{ km}}$

(b) For Mt. Everest, 29 300 ft = 8.88 km, $\boxed{\text{Yes}}$.

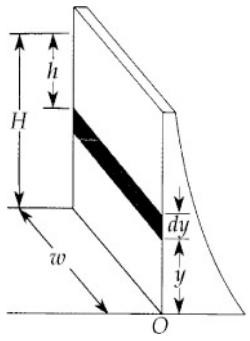
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P14.70 (a) The torque is

$$\tau = \int d\tau = \int r dF$$

From ANS. FIG. P14.70,

$$\tau = \int_0^H y\left[\rho g(H-y)w dy\right] = \boxed{\frac{1}{6}\rho g w H^3}$$



ANS. FIG. P14.70

(b) The total force is given as $\frac{1}{2}\rho g w H^2$.

If this were applied at a height y_{eff} such that the torque remains unchanged, we have

$$\frac{1}{6}\rho g w H^3 = y_{eff}\left[\frac{1}{2}\rho g w H^2\right] \quad \text{and} \quad y_{eff} = \boxed{\frac{1}{3}H}$$

P14.71 Looking first at the top scale and the iron block, we have

$$T + B = F_{g,\,\text{iron}}$$

where T is the tension in the spring scale, B is the buoyant force, and $F_{g,\,\text{iron}}$ is the weight of the iron block. Now if m_{iron} is the mass of the iron block, we have

$$m_{\text{iron}} = \rho_{\text{iron}} V$$

so $$V = \frac{m_{\text{iron}}}{\rho_{\text{iron}}} = V_{\text{displaced oil}}$$

Then, $$B = \rho_{\text{oil}} V_{\text{iron}} g$$

Therefore,

$$T = F_{g,\,\text{iron}} - \rho_{\text{oil}} V_{\text{iron}} g = m_{\text{iron}} g - \rho_{\text{oil}} \frac{m_{\text{iron}}}{\rho_{\text{iron}}}$$

or

$$T = \left(1 - \frac{\rho_{\text{oil}}}{\rho_{\text{iron}}}\right) m_{\text{iron}} g$$

$$= \left(1 - \frac{916\ \text{kg/m}^3}{7\,860\ \text{kg/m}^3}\right)(2.00\ \text{kg})(9.80\ \text{m/s}^2)$$

$$= \boxed{17.3\ \text{N}}$$

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Next, we look at the bottom scale which reads n (i.e., exerts an upward normal force n on the system). Consider the external vertical forces acting on the beaker-oil-iron combination.

$$\sum F_y = 0 \text{ gives}$$

$$T + n - F_{g,\text{beaker}} - F_{g,\text{oil}} - F_{g,\text{iron}} = 0 \rightarrow$$

$$n = \left(m_{\text{beaker}+} m_{\text{oil}+} m_{\text{iron}}\right) g - T = \left(5.00 \text{ kg}\right)\left(9.80 \text{ m/s}^2\right) - 17.3 \text{ N}$$

Thus, $\boxed{n = 31.7 \text{ N}}$ is the lower scale reading.

P14.72 Looking at the top scale and the iron block:

$$T + B = F_{g,\text{ Fe}}, \qquad \text{where} \qquad B = \rho_o V_{\text{Fe}} g = \rho_o \left(\frac{m_{\text{Fe}}}{\rho_{\text{Fe}}}\right) g$$

is the buoyant force exerted on the iron block by the oil.

Thus, $T = F_{g,\text{ Fe}} - B = m_{\text{Fe}} g - \rho_o \left(\frac{m_{\text{Fe}}}{\rho_{\text{Fe}}}\right) g,$

or $\boxed{T = \left(1 - \frac{\rho_o}{\rho_{\text{Fe}}}\right) m_{\text{Fe}} g}$ is the reading on the top scale.

Now, consider the bottom scale, which exerts an upward force of n on the beaker-oil-iron combination.

$$\sum F_y = 0:$$

$$T + n - F_{g,\text{beaker}} - F_{g,\text{oil}} - F_{g,\text{Fe}} = 0$$

$$n = (m_b + m_o + m_{\text{Fe}}) g - T$$

$$n = (m_b + m_o + m_{\text{Fe}}) g - \left(1 - \frac{\rho_o}{\rho_{\text{Fe}}}\right) m_{\text{Fe}} g$$

or the reading of the bottom scale is

$$\boxed{n = \left[m_b + m_o + \left(\frac{\rho_o}{\rho_{\text{Fe}}}\right) m_{\text{Fe}}\right] g}$$

P14.73 Let f represent the fraction of the volume V occupied by zinc in the new coin. We have $m = \rho V$ for both coins:

$$3.083 \text{ g} = (8.920 \text{ g/cm}^3) V$$

and $$2.517 \text{ g} = \left(7.133 \text{ g/cm}^3\right)(fV) + \left(8.920 \text{ g/cm}^3\right)(1 - f)V$$

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By substitution,

$$2.517\ \text{g} = (7.133\ \text{g/cm}^3)fV + 3.083\ \text{g} - (8.920\ \text{g/cm}^3)fV$$

$$fV = \frac{3.083\ \text{g} - 2.517\ \text{g}}{8.920\ \text{g/cm}^3 - 7.133\ \text{g/cm}^3}$$

and again substituting to eliminate the volume,

$$f = \frac{0.566\ \text{g}}{1.787\ \text{g/cm}^3}\left(\frac{8.920\ \text{g/cm}^3}{3.083\ \text{g}}\right) = 0.916\ 4 = \boxed{91.64\%}$$

***P14.74** Let ℓ represent the length below water at equilibrium and M the tube's mass:

$$\sum F_y = 0 \qquad \text{gives} \qquad -Mg + \rho\pi r^2 \ell g = 0$$

Now with any excursion x from equilibrium:

$$-Mg + \rho\pi r^2 (\ell - x) g = Ma$$

Subtracting the equilibrium equation gives:

$$-\rho\pi r^2 g x = Ma$$

$$a = -\left(\frac{\rho\pi r^2 g}{M}\right)x = -\omega^2 x$$

The opposite direction and direct proportionality of a to x imply SHM with angular frequency

$$\omega = \sqrt{\frac{\rho\pi r^2 g}{M}}$$

$$T = \frac{2\pi}{\omega} = \boxed{\frac{2}{r}\sqrt{\frac{\pi M}{\rho g}}}$$

P14.75 Pascal's principle, $\frac{F_1}{A_1} = \frac{F_2}{A_2}$, or $\frac{F_{\text{pedal}}}{A_{\substack{\text{Master}\\\text{cylinder}}}} = \frac{F_{\text{brake}}}{A_{\substack{\text{brake}\\\text{cylinder}}}}$, gives

$$F_{\text{brake}} = \left(\frac{A_{\text{brake cylinder}}}{A_{\text{master cylinder}}}\right)F_{\text{pedal}} = \left(\frac{6.4\ \text{cm}^2}{1.8\ \text{cm}^2}\right)(44\ \text{N}) = 156\ \text{N}$$

This is the normal force exerted on the brake shoe. The frictional force is

$$f = \mu_k n = 0.50(156\ \text{N}) = 78\ \text{N}$$

and the torque is $\tau = f \cdot r_{\text{drum}} = (78\ \text{N})(0.34\ \text{m}) = \boxed{27\ \text{N}\cdot\text{m}}$.

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***P14.76** (a) Since the upward buoyant force is balanced by the weight of the sphere,

$$m_1 g = \rho V g = \rho\left(\frac{4}{3}\pi R^3\right)g$$

In this problem, $\rho = 0.789\ 45\ \text{g/cm}^3$ at 20.0°C, and $R = 1.00$ cm, so we find

$$m_1 = \rho\left(\frac{4}{3}\pi R^3\right) = \left(0.789\ 45\ \text{g/cm}^3\right)\left[\frac{4}{3}\pi(1.00\ \text{cm})^3\right]$$

$$= \boxed{3.307\ \text{g}}$$

(b) Following the same procedure as in part (a), with $\rho' = 0.780\ 97\ \text{g/cm}^3$ at 30.0°C, we find

$$m_2 = \rho'\left(\frac{4}{3}\pi R^3\right) = \left(0.780\ 97\ \text{g/cm}^3\right)\left[\frac{4}{3}\pi(1.00\ \text{cm})^3\right]$$

$$= \boxed{3.271\ \text{g}}$$

(c) When the first sphere is resting on the bottom of the tube, $n + B = F_{g1} = m_1 g$, where n is the normal force.

Since $B = \rho' V g$,

$$n = m_1 g - \rho' V g$$

$$= \left[3.307\ \text{g} - \left(0.780\ 97\ \text{g/cm}^3\right)\frac{4}{3}\pi(1.00\ \text{cm})^3\right]\left(980\ \text{cm/s}^2\right)$$

$$n = 34.8\ \text{g}\cdot\text{cm/s}^2 = \boxed{3.48\times10^{-4}\ \text{N}}$$

P14.77 The disk (mass $M = 10.0$ kg, radius $R = 0.250$ m) has moment of inertia $I = \frac{1}{2}MR^2$. The disk slows from $\omega_i = 300$ rev/min to $\omega_f = 0$ in time interval $\Delta t = 60.0$ s. Its angular acceleration is

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{\omega_f - \omega_i}{\Delta t} = \frac{-\omega_i}{\Delta t}$$

Frictional torque from the brake pad slows the wheel. Friction has moment arm $d = 0.220$ m. The relation between friction and angular acceleration is

$$\sum\tau = I\alpha: \quad -fd = I\alpha \quad \rightarrow \quad f = -\frac{I}{d}\alpha = -\frac{\frac{1}{2}MR^2}{d}\left(\frac{-\omega_i}{\Delta t}\right)$$

$$\rightarrow f = \frac{MR^2\omega_i}{2d\Delta t}$$

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The normal force and coefficient of friction $(\mu_k = 0.500)$ between the brake pad and the disk determine the amount of friction. We can write an expression for the normal force:

$$f = \mu_k n \quad \rightarrow \quad n = \frac{f}{\mu_k} = \frac{MR^2\omega_i}{2\mu_k d\Delta t}$$

The pressure of the brake fluid acting on a piston of area A (diameter $D = 5.00$ cm, radius $r = D/2 = 0.0250$ m) produces the normal force that the brake pad exerts on the disk. The pressure in the brake fluid is

$$P = \frac{n}{\pi r^2} = \frac{MR^2\omega_i}{\left(2\mu_k d\Delta t\right)\pi r^2}$$

$$P = \frac{(10.0 \text{ kg})(0.250 \text{ m})^2\left[\left(\frac{300 \text{ rev}}{\text{min}}\right)\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right)\left(\frac{1 \text{ min}}{60.0 \text{ s}}\right)\right]}{2(0.500)(0.220 \text{ m})(60.0 \text{ s})\pi(0.0250 \text{ m})^2}$$

$$= \boxed{758 \text{ Pa}}$$

P14.78 (a) Since the pistol is fired horizontally, the emerging water stream has initial velocity components of $(v_{0x} = v_{\text{nozzle}}, v_{0y} = 0)$. Then, $\Delta y = v_{0y}t + \frac{1}{2}a_y t^2$, with $a_y = -g$, gives the time of flight as

$$t = \sqrt{\frac{2\left(\Delta y\right)}{a_y}} = \sqrt{\frac{2\left(-1.50 \text{ m}\right)}{-9.80 \text{ m/s}^2}} = \boxed{0.553 \text{ s}}$$

(b) With $a_x = 0$, and $v_{0x} = v_{\text{nozzle}}$, the horizontal range of the emergent stream is $\Delta x = v_{\text{nozzle}}t$, where t is the time of flight from above. Thus, the speed of the water emerging from the nozzle is

$$v_{\text{nozzle}} = \frac{\Delta x}{t} = \frac{8.00 \text{ m}}{0.553 \text{ s}} = \boxed{14.5 \text{ m/s}}$$

(c) From the equation of continuity, $A_1 v_1 = A_2 v_2$, the speed of the water in the larger cylinder is $v_1 = (A_2/A_1)v_2 = (A_2/A_1)v_{\text{nozzle}}$, or

$$v_1 = \left(\frac{\pi r_2^2}{\pi r_1^2}\right)v_{\text{nozzle}} = \left(\frac{r_2}{r_1}\right)^2 v_{\text{nozzle}} = \left(\frac{1.00 \text{ mm}}{10.0 \text{ mm}}\right)^2\left(14.5 \text{ m/s}\right)$$

$$= \boxed{0.145 \text{ m/s}}$$

(d) The pressure at the nozzle is atmospheric pressure, or $\boxed{P_2 = 1.013 \times 10^5 \text{ Pa}}$.

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(e) With the two cylinders horizontal, $y_1 = y_2$ and gravity terms from Bernoulli's equation can be neglected, leaving

$$P_1 + \frac{1}{2}\rho_w v_1^2 = P_2 + \frac{1}{2}\rho_w v_2^2$$

so the needed pressure in the larger cylinder is

$$\begin{aligned} P_1 &= P_2 + \frac{\rho_w}{2}\left(v_2^2 - v_1^2\right) \\ &= 1.013\times10^5 \text{ Pa} \\ &\quad + \frac{1.00\times10^3 \text{ kg/m}^3}{2}\left[(14.5 \text{ m/s})^2 - (0.145 \text{ m/s})^2\right] \end{aligned}$$

or

$$P_1 = \boxed{2.06\times10^5 \text{ Pa}}$$

(f) To create an overpressure of $\Delta P = 2.06\times10^5 \text{ Pa} = 1.05\times10^5 \text{ Pa}$ in the larger cylinder, the force that must be exerted on the piston is

$$\begin{aligned} F_1 &= (\Delta P)A_1 = (\Delta P)(\pi r_1^2) \\ &= (1.05\times10^5 \text{ Pa})\pi(1.00\times10^{-2} \text{ m})^2 \\ &= \boxed{33.0 \text{ N}} \end{aligned}$$

P14.79 Energy for the fluid-Earth system is conserved.

$$\begin{aligned} (K+U)_i &= (K+U)_f \\ 0 + \frac{mgL}{2} + 0 &= \frac{1}{2}mv^2 + 0 \\ v &= \sqrt{gL} = \sqrt{(2.00 \text{ m})(9.80 \text{ m/s}^2)} = \boxed{4.43 \text{ m/s}} \end{aligned}$$

P14.80 (a) The flow rate, Av, as given may be expressed as follows:

$$\frac{25.0 \text{ liters}}{30.0 \text{ s}} = 0.833 \text{ liters/s} = 833 \text{ cm}^3/\text{s}$$

The area of the faucet tap is $\pi \text{ cm}^2$, so we can find the velocity as

$$v = \frac{\text{flow rate}}{A} = \frac{833 \text{ cm}^3/\text{s}}{\pi \text{ cm}^2} = 265 \text{ cm/s} = \boxed{2.65 \text{ m/s}}$$

(b) We choose point 1 to be in the entrance pipe and point 2 to be at the faucet tap. $A_1 v_1 = A_2 v_2$ gives $v_1 = 0.295$ m/s. Bernoulli's equation is:

$$P_1 - P_2 = \frac{1}{2}\rho\left(v_2^2 - v_1^2\right) + \rho g\left(y_2 - y_1\right)$$

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and gives

$$P_1 - P_2 = \frac{1}{2}\left(10^3 \text{ kg/m}^3\right)\left[(2.65 \text{ m/s})^2 - (0.295 \text{ m/s})^2\right] + \left(10^3 \text{ kg/m}^3\right)(9.80 \text{ m/s})(2.00 \text{ m})$$

or $P_{\text{gauge}} = P_1 - P_2 = \boxed{2.31 \times 10^4 \text{ Pa}}$.

P14.81 (a) Consider the pressure at points A and B in ANS. FIG. P14.81(b).

Using the left tube: $P_A = P_{\text{atm}} + \rho_w g(L - h)$

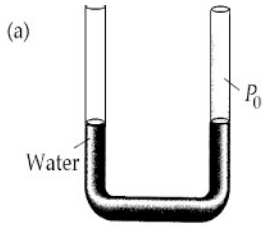
Using the right tube: $P_B = P_{\text{atm}} + \rho_o g L$

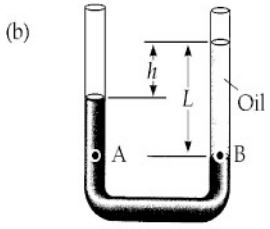
But Pascal's principle says that $P_A = P_B$.

Therefore, $P_{\text{atm}} + \rho_w g(L - h) = P_{\text{atm}} + \rho_o g L$

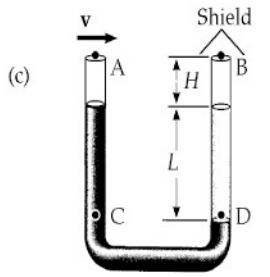
or $\rho_w h = (\rho_w - \rho_o)L$, giving

$$h = \left(\frac{\rho_w - \rho_o}{\rho_w}\right)L = \left(\frac{1\,000 \text{ kg/m}^3 - 750 \text{ kg/m}^3}{1\,000 \text{ kg/m}^3}\right)(5.00 \text{ cm}) = \boxed{1.25 \text{ cm}}$$





(b) Consider part (c) of the diagram showing the situation when the air flow over the left tube equalizes the fluid levels in the two tubes. First, apply Bernoulli's equation to points A and B ($y_A = y_B$, $v_A = v$, and $v_B = 0$).



ANS. FIG. P14.81

This gives:

$$P_A + \frac{1}{2}\rho_a v^2 + \rho_a g y_A = P_B + \frac{1}{2}\rho_a (0)^2 + \rho_a g y_B$$

and since $y_A = y_B$, this reduces to

$$P_B - P_A = \frac{1}{2}\rho_a v^2 \qquad \textbf{[1]}$$

Now consider points C and D, both at the level of the oil-water interface in the right tube. Using the variation of pressure with

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depth in static fluids, we have

$$P_C = P_A + \rho_a g H + \rho_w g L$$

and $$P_D = P_B + \rho_a g H + \rho_o g L$$

But Pascal's principle says that $P_C = P_D$. Equating these two gives:

$$P_B + \rho_a g H + \rho_o g L = P_A + \rho_a g H + \rho_w g L$$

or $$P_B - P_A = (\rho_w - \rho_o) g L \qquad \textbf{[2]}$$

Substitute equation [1] for $P_B - P_A$ into [2] to obtain

$$\frac{1}{2}\rho_a v^2 = (\rho_w - \rho_o) g L$$

or

$$v = \sqrt{\frac{2gL(\rho_w - \rho_o)}{\rho_a}}$$

$$= \sqrt{2(9.80\ \text{m/s}^2)(0.050\ 0\ \text{m})\left(\frac{1\ 000\ \text{kg/m}^3 - 750\ \text{kg/m}^3}{1.20\ \text{kg/m}^3}\right)}$$

$$v = \boxed{14.3\ \text{m/s}}$$

P14.82 Take point ① at the free water surface in the tank and point ② at the bottom end of the tube:

$$P_1 + \rho g y_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2}\rho v_2^2$$

$$P_0 + \rho g d + 0 = P_0 + 0 + \frac{1}{2}\rho v_2^2$$

$$v_2 = \sqrt{2gd}$$

The volume flow rate is $\frac{V}{t} = \frac{Ah}{t} = v_2 A'$. Then $t = \frac{Ah}{v_2 A'} = \frac{Ah}{A'\sqrt{2gd}}$.

P14.83 (a) For diverging streamlines that pass just above and just below the hydrofoil, we have

$$P_t + \rho g y_t + \frac{1}{2}\rho v_t^2 = P_b + \rho g y_b + \frac{1}{2}\rho v_b^2$$

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Ignoring the buoyant force means taking $y_t \approx y_b$:

$$P_t + \frac{1}{2}\rho\left(nv_b\right)^2 = P_b + \frac{1}{2}\rho v_b^2$$

$$P_b - P_t = \frac{1}{2}\rho v_b^2\left(n^2 - 1\right)$$

The lift force is $\left(P_b - P_t\right)A = \frac{1}{2}\rho v_b^2\left(n^2 - 1\right)A.$

(b) For liftoff,

$$\frac{1}{2}\rho v_b^2\left(n^2 - 1\right)A = Mg$$

$$v_b = \left(\frac{2Mg}{\rho\left(n^2 - 1\right)A}\right)^{1/2}$$

The speed of the boat relative to the shore must be nearly equal to this speed of the water below the hydrofoil relative to the boat.

***P14.84** First, consider the path from the viewpoint of projectile motion to find the speed at which the water emerges from the tank. From $\Delta y = v_{yi}t + \frac{1}{2}a_y t^2$ with $v_{yi} = 0$, $\Delta y = -1.00$ m, and $a_y = -g$, we find the time of flight as

$$t = \sqrt{\frac{2\left(\Delta y\right)}{a_y}} = \sqrt{\frac{2.00\text{ m}}{g}} = 0.452\text{ s}$$

From the horizontal motion, the speed of the water coming out of the hole is

$$v_2 = v_{xi} = \frac{\Delta x}{t} = \frac{0.600\text{ m}}{0.452\text{ s}} = 1.33\text{ m/s}$$

We now use Bernoulli's equation, with point 1 at the top of the tank and point 2 at the level of the hole. With $P_1 = P_1 = P_{\text{atm}}$ and $v_1 \approx 0$, this gives

$$\rho g y_1 = \rho g y_2 + \frac{1}{2}\rho v_2^2$$

or

$$h = y_1 - y_2 = \frac{v_2^2}{2g} = \frac{\left(1.33\text{ m/s}\right)^2}{2g} = 9.00\times10^{-2}\text{ m} = \boxed{9.00\text{ cm}}$$

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Challenge Problems

P14.85 Let s stand for the edge of the cube, h for the depth of immersion, ρ_{ice} for the density of the ice, ρ_{w} for the density of water, and ρ_{al} for the density of the alcohol.

(a) According to Archimedes's principle, at equilibrium we have

$$B = F_{\text{g}}$$

$$\rho_{\text{w}} g h s^2 = \rho_{\text{ice}} g s^3 \Rightarrow h = s\frac{\rho_{\text{ice}}}{\rho_{\text{w}}}$$

With $\rho_{\text{ice}} = 0.917 \times 10^3 \text{ kg/m}^3$,

$$\rho_{\text{w}} = 1.00 \times 10^3 \text{ kg/m}^3$$

and $s = 20.0$ mm,

we obtain $h = 20.0(0.917) = 18.34 \text{ mm} \approx \boxed{18.3 \text{ mm}}$

(b) We assume that the top of the cube is still above the alcohol surface. Letting h_{al} stand for the thickness of the alcohol layer, we have

$$\rho_{\text{al}} g s^2 h_{\text{al}} + \rho_{\text{w}} g s^2 h_{\text{w}} = \rho_{\text{ice}} g s^3 \quad \text{so} \quad h_{\text{w}} = \left(\frac{\rho_{\text{ice}}}{\rho_{\text{w}}}\right) s - \left(\frac{\rho_{\text{al}}}{\rho_{\text{w}}}\right) h_{\text{al}}.$$

With $\rho_{\text{al}} = 0.806 \times 10^3 \text{ kg/m}^3$

and $h_{\text{al}} = 5.00$ mm,

we obtain $h_{\text{w}} = 18.34 - 0.806(5.00) = 14.31 \text{ mm} \approx \boxed{14.3 \text{ mm}}$.

To check our assumption above, the bottom of the cube is below the top surface of the alcohol 14.4 mm + 5.00 mm = 19.3 mm, so the top of the cube is above the surface of the alcohol 20.0 mm – 19.3 mm = 0.7 mm. The assumption was valid.

(c) Here , $h'_{\text{w}} = s - h'_{\text{al}}$, so Archimedes's principle gives

$$\rho_{\text{al}} g s^2 h'_{\text{al}} + \rho_{\text{w}} g s^2 (s - h'_{\text{al}}) = \rho_{\text{ice}} g s^3 \Rightarrow \rho_{\text{al}} h'_{\text{al}} + \rho_{\text{w}} (s - h'_{\text{al}}) = \rho_{\text{ice}} s$$

$$h'_{\text{al}} = s\frac{(\rho_{\text{w}} - \rho_{\text{ice}})}{(\rho_{\text{w}} - \rho_{\text{al}})} = (20.0 \text{ mm})\frac{(1.000 - 0.917)}{(1.000 - 0.806)}$$

$$= 8.557 \approx \boxed{8.56 \text{ mm}}$$

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P14.86 Assume the top of the barge without the pile of iron has height H_0 above the surface of the water. When a mass of iron M_{Fe} is added to the barge, the barge sinks a distance ΔH until the buoyant force from the water equals the additional weight of the iron. The barge is a square with sides of length L, so the volume of displaced water is $L^2\Delta H$, and the buoyant force supporting the extra weight is

$$B = \left(\rho_w L^2 \Delta H\right) g = M_{Fe} g$$

where ρ_w is the density of water.

The scrap iron pile has the shape of a cone, and the volume of a cone of base radius R and central height h is $V_{cone} = \pi R^2 h/3$; therefore, the mass of the iron is $M_{Fe} = \rho_{Fe}\pi R^2 h/3$, where ρ_{Fe} is the density of iron. We find the distance the barge sinks with a pile of iron:

$$B = \left(\rho_w L^2 \Delta H\right) g = M_{Fe} g$$

$$\left(\rho_w L^2 \Delta H\right) g = \left(\rho_{Fe}\pi R^2 h/3\right) g \rightarrow \Delta H = \left(\frac{\rho_{Fe}}{\rho_w}\right)\left(\frac{\pi}{3}\right)\left(\frac{R^2}{L^2}\right) h$$

If the iron is piled to a height h, the barge will sink by the distance ΔH, so the distance from the water level to the top of the iron pile is $D_{top} = H_0 - \Delta H + h$.

For the situation of the problem, side $L = 2r$, and the initial conical pile of scrap iron has radius $R = r$ and height is $h = r$. The distance the barge sinks is

$$\Delta H = \left(\frac{\rho_{Fe}}{\rho_w}\right)\left(\frac{\pi}{3}\right)\left(\frac{R^2}{L^2}\right) h$$

$$\Delta H = \left(\frac{\rho_{Fe}}{\rho_w}\right)\left(\frac{\pi}{3}\right)\left(\frac{r^2}{(2r)^2}\right) r = \left(\frac{\rho_{Fe}}{\rho_w}\right)\left(\frac{\pi}{3}\right)\left(\frac{r^2}{4r^2}\right) r = \left(\frac{\rho_{Fe}}{\rho_w}\right)\left(\frac{\pi}{12}\right) r$$

and the height of the top of the pile above the water is

$$D_{top} = H_0 - \Delta H + h = H_0 - \left(\frac{\rho_{Fe}}{\rho_w}\right)\left(\frac{\pi}{12}\right) r + r$$

For $\rho_w = 1.00\times 10^3$ kg/m^3 and $\rho_{Fe} = 7.86\times 10^3$ kg/m^3, this expression becomes

$$D_{top} = H_0 - \left(\frac{7.86\times 10^3}{1.00\times 10^3}\right)\left(\frac{\pi}{12}\right) r + r = H_0 - 2.06r + r$$

$$D_{top} = H_0 - 1.06r$$

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This distance is too large to allow the barge to go under the bridge:

$$D_{\text{top}} = H_0 - 1.06r \geq D_{\text{bridge}}$$

When the pile is reduced to a height h', but still with the same base radius $R = r$, the distance the barge sinks is

$$\Delta H = \left(\frac{\rho_{\text{Fe}}}{\rho_{\text{w}}}\right)\left(\frac{\pi}{12}\right)h' = 2.06h'$$

The height of the top of the pile above the water is now

$$D'_{\text{top}} = H_0 - \Delta H + h' = H_0 - 2.06h' + h' = H_0 - 1.06h'$$

but this means the top of the pile is now higher! To check this, recall that the height of the pile is reduced, so $h' < r$:

$$D'_{\text{top}} > D_{\text{top}}$$

$$H_0 - 1.06h' > H_0 - 1.06r \quad \rightarrow \quad -1.06h' > -1.06r \quad \rightarrow \quad h' < r$$

which is true.

The situation is impossible because lowering the height of the iron pile on the barge while keeping the base radius the same results in the top of the pile rising higher above the water level.

P14.87 The incremental version of $P - P_0 = \rho g y$ is $dP = -\rho g dy$.

We assume that the density of air is proportional to pressure,

or $\frac{P}{\rho} = \frac{P_0}{\rho_0}$. Combining these two equations we have

$$dP = -P\frac{\rho_0}{P_0}g dy$$

Integrating both sides,

$$\int_{P_0}^{P}\frac{dP}{P} = -g\frac{\rho_0}{P_0}\int_0^y dy$$

gives $$\ln\left(\frac{P}{P_0}\right) = -\frac{\rho_0 g y}{P_0}$$

Defining $\alpha = \frac{\rho_0 g}{P_0}$ then gives $P = P_0 e^{-\alpha y}$.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P14.2 (a) ~4×10^{17} kg/m^3; (b) See P14.2 for the full description.

P14.4 5.27×10^{18} kg

P14.6 (a) 65.1 N; (b) 275 N

P14.8 225 N

P14.10 (a) 5.88×10^6 N down; (b) 196 kN outward; (c) 588 kN outward

P14.12 The situation is impossible because the longest straw Superman can use and still get a drink is less than 12.0 m.

P14.14 1.05×10^5 Pa

P14.16 (a) $F = \frac{1}{2}\rho g w h\left(2d - h\right)$; (b) $\tau = \frac{1}{2}\rho g h\left(dh^2 - \frac{1}{3}h^3\right)$

P14.18 0.072 1 mm

P14.20 (a) 14.7 kPa, 0.015 5 atm, 11.8 m; (b) Blockage of the fluid within the spinal column or between the skull and the spinal column would prevent the fluid level from rising.

P14.22 (a) 20.0 cm; (b) 0.490 cm

P14.24 (a) $P = P_0 + \rho g h$; (b) Mg/A

P14.26 3.33×10^3 kg/m^3

P14.28 (a) 444 kg; (b) 480 kg

P14.30 1.28×10^4 m^2

P14.32 2.67×10^3 kg

P14.34 (a) B = 25.0 N; (b) horizontally inward; (c) The string tension increases. The water under the block pushes up on the block more strongly than before because the water is under higher pressure due to the weight of the oil above it; (d) 62.5%

P14.36 See P14.36 for the full derivation.

P14.38 (a) 3.7 kN; (b) 1.9 kN; (c) Atmospheric pressure at this high altitude is much lower than at the Earth's surface

P14.40 (a) 0.471 m/s; (b) 4.24 m/s

P14.42 (a) 0.638 m/s; (b) 2.55 m/s; (c) 1.80×10^{-3} m^3/s

P14.44 (a) 27.9 N; (b) 3.32×10^4 kg; (c) 7.26×10^4 Pa

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P14.46 (a) $P = \dfrac{\Delta E}{\Delta t} = \dfrac{\Delta mgh}{\Delta t} = \left(\dfrac{\Delta m}{\Delta t}\right)gh = Rgh$; (b) 616 MW

P14.48 (a) $\left(3.93 \times 10^{-6}\ \text{m}^3/\text{s}\right)\sqrt{\Delta P}$ where ΔP is in pascal; (b) 0.305 L/s; (c) 0.431 L/s

P14.50 (a) 28.0 m/s; (b) 28.0 m/s; (c) The answers agree precisely. The models are consistent with each other. (d) 2.11 MPa

P14.52 (a) 6.80×10^4 Pa; (b) Higher. With the inclusion of another upward force due to deflection of air downward, the pressure difference does not need to be as great to keep the airplane in flight.

P14.54 (a) 452 N outward; (b) 1.81 kN outward

P14.56 1.01 kJ

P14.58 $\left(P_0 - P\right)\pi R^2$

P14.60 (a) A particle in equilibrium model; (b) $\sum F_y = B - F_b - F_{\text{He}} - F_s = 0$; (c) $m_s = \left(\rho_{\text{air}} - \rho_{\text{He}}\right)\dfrac{4}{3}\pi r^2 - m_b$; (d) 0.023 7 kg; (e) 0.948 m

P14.62 See P14.62 for full description.

P14.64 (a) 8.04 m/s; (b) The gravitational force and the buoyant force; (c) The net upward force on the ball brings it downward motion to a stop, 4.18 m; (d) 8.04 m/s; (e) The time intervals are equal; (f) See P14.64(f) for a full conceptual argument.

P14.66 $\sim 10^4$

P14.68 (a) See P14.68(a) for full description; (b) 2.66×10^{-3} m; (c) The situation in the human body is not represented by a large artery feeding into a single capillary as in part (b). See P14.68(c) for full explanation.

P14.70 (a) $\dfrac{1}{6}\rho g w H^3$; (b) $\dfrac{1}{3}H$

P14.72 $T = \left(1 - \dfrac{\rho_{\text{o}}}{\rho_{\text{Fe}}}\right)m_{\text{Fe}}g,\ n = \left[m_b + m_{\text{o}} + \left(\dfrac{\rho_{\text{o}}}{\rho_{\text{Fe}}}\right)\right]g$

P14.74 $\dfrac{2}{r}\sqrt{\dfrac{\pi M}{\rho g}}$

P14.76 (a) 3.307 g; (b) 3.271 g; (c) 3.48×10^{-4} N

P14.78 (a) 0.553 s; (b) 14.5 m/s; (c) 0.145 m/s; (d) $P_2 = 1.013 \times 10^5$ Pa; (e) 2.06×10^5 Pa; (f) 33.0 N

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P14.80 (a) 2.65 m/s; (b) 2.31×10^4 Pa

P14.82 See P14.82 for the full answer.

P14.84 9.00 cm

P14.86 The situation is impossible because lowering the height of the iron pile on the barge while keeping the base radius the same results in the top of the pile rising higher above the water level.

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